Repair Manual

911 Carrera
(993)

Volume III:
Transmission
Automatic

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Preface

Structure

The "Technical Literture" for the "911 Carrera (993)" model is basically structured as before, i.e. the structure follows the familiar repair groups.

A new feature is that the structure includes the main groups **0 to 9** and the main group **D**.

Main groups:	0	Complete vehicle – General
	1	Engine
	2	Fuel, exhaust, engine electrical system
	3	Transmission
	4	Chassis
	5	Body
	6	Body equipment, outside
	7	Body equipment, interior
	8	Air conditioning
	9	Electrical system
	D	Diagnosis

Layout

The layout in the below items remains unchanged throughout the repair manual

1. Table of tightening torques
2. Special tools required
3. Exploded views
4. Legends for the exploded views
5. Assembly notes / use of special tools

As a new feature, however, the former item 6 (Repair group diagnosis) is no longer filed in the volume corresponding to the respective repair group. The **Diagnosis test plans / diagnosis procedures** have been combined in a **separate Diagnosis volume** broken down according to the main groups 0 to 9.

Another new feature is that the contents of the "Service Information Technik" are indicated in the Repair Manual. This brochure concentrates on a description of the design and function of components and of the new features introduced for a particular model year.

Service Number

All major repair procedures and repair descriptions are identified by a two- or four-digit **Service Number** completed by two additional digits to identify the work that corresponds to the first six digits of the working position number in the Working Times and Damage Catalog.

Example: 30 37 37 Dismantling and assembling clutch control shaft

Explanation: 30 37 37 50 (full working position number)

Repair group
here: Clutch, control

Component designation
here: Clutch control shaft

Activity
here: Dismantling and assembling

Index
here: Removed

Presentation in the various documents

30 37 37 50	Working position no. from **Working Times and Damage Catalog**, consisting of repair group, component designation, activity and index
30 37 37	Six-digit number in **Repair Manual**, consisting of repair group, component designation and activity
	Service number in **Service Information**, consisting of repair group and component designation

Goal

The introduction of a service number in the "technical literature" is intended to facilitate standardization and positive identification to allow direct cross-referencing among the various documents. This is of particular importance with regard to the use of electronic media.

Survey of contents of Service Information Technik '95

The Service Information gives a detailed description of the technical features of the new 911 Carrera.

Electrical system

Summary

III Transmission Automatic Transmission

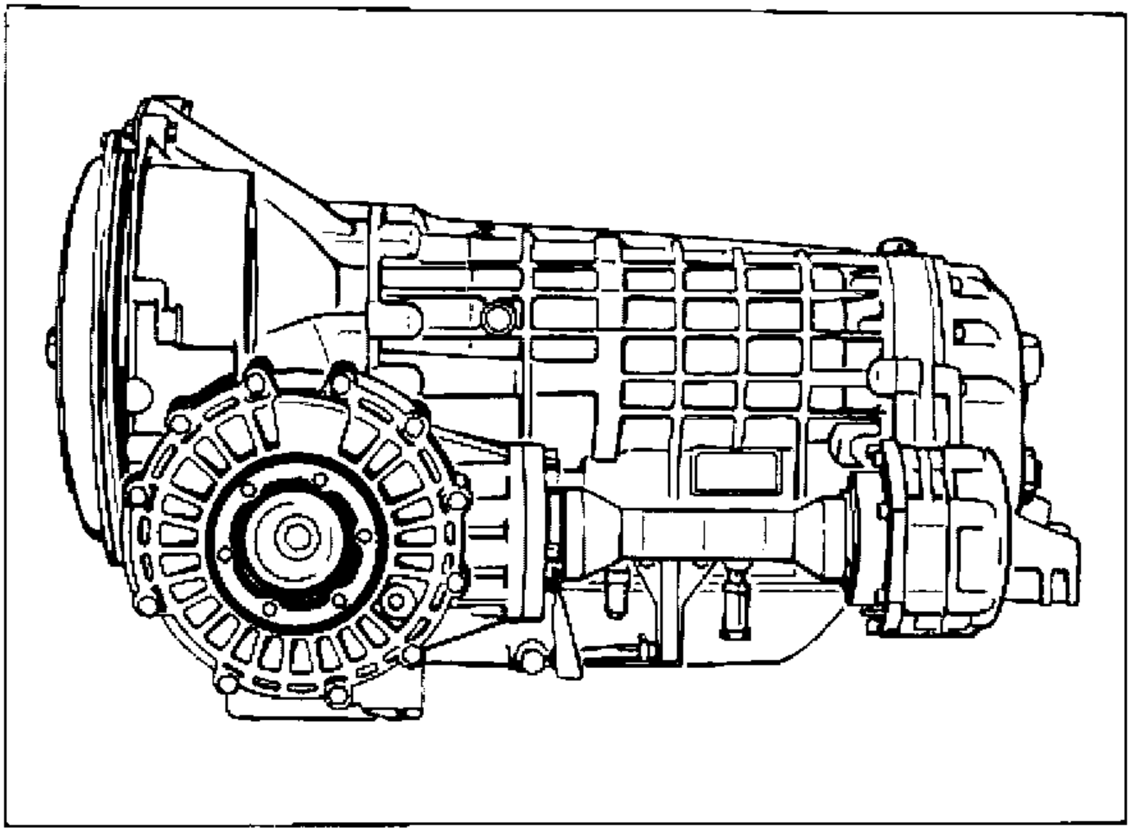
Printed in Germany - 13, 1995

39 Final drive, differential, differential lock

Printed in Germany - 13, 1995

3 Technical Data

4-speed Tiptronic transmission A50



Type	Code letter	Version	Installed in	Model year
A 50/04		4-speed	911 Carrera worldwide except USA and Taiwan	'94 / '95
A 50/05		4-speed	911 Carrera USA, Taiwan	'94 / '95

3 Technical Data

General data	A50/04	A50/05
Type	fully automatic 4-speed planetary transmission (Tiptronic)	
Gear ratios		
spur gear	1.100	1.100
1st gear	2.479	2.479
2nd gear	1.479	1.479
3rd gear	1.000	1.000
4th gear	0.728	0.728
reverse	2.086	2.086
final drive	hypoid bevel gear with 15 mm offset	
final drive ratio	9:33 = 3.667	9 : 32 = 3.556
stall speed	2300 - 400	2300 - 400
oil volume for final drive		approx. 0.9 l multi-grade transmission oil 75 W 90 API specification GL5 (**MIL-L 2105 B**), or SAE 90
oil volume for automatic transmission with torque converter		total volume approx. 9.5 l oil change approx. 3.5 l ATF-Dexron II D

3 Technical Data

Torque specifications for Tiptronic transmission

Location	Thread	Tightening Torque Nm (ftlb.)
Multifunctional switch to transmission	M 6 x 25	10 (7)
Selector lever to selector shaft	M 8 x 1	15 (11)
Long halfshaft flange to transmission housing	M 8	23 (17)
Short halfshaft flange to differential	M 10 x 60	46 (34)
Plug to rear transmission housing	M 22 x 1.5	50 (37)
Rear transmission housing to automatic transmission	M 10	46 (34)
Front transmission cover to intermediate plate	M 10 x 35 M 8	46 (34) 23 (17)
Intermediate plate to automatic transmission	M 8	23 (17)
Drive pinion bearing assembly to front transmission cover	M 8	23 (17)
Fastening nut to helical gear	M 40 x 1.5	250 (184)
Guide part for parking lock to housing	M 6 x 20	10 (7)
Plug to ATF pan	M 14 x 1.5	40 (30)
Banjo bolt to ATF pan	M 12 x 1.5	40 (30)
Banjo bolt to housing	M 14 x 1.5	40 (30)
ATF pan to housing	M 6	6 (4)

Location	Thread	Tightening torque Nm *(ftlb.)*
ATF strainer to hydraulic control unit	M 6 x 65	
Plug to ATF quick-fill adapter	M 14 x 1.5	30 (22)
Hydraulic control unit to transmission	M 6	
Hexagon nut for transmission socket		12 (9)
ATF indicator tube to transmission	M 6 x 4	
Adapter of hydraulic control unit to transmission housing	M 6	
Pressure regulator and solenoid valve mount to control unit	M 6	
Solenoid valves to hydraulic control unit	M 5 x 12	
ATF pump to housing	M 6	
Oil drainage and filling plug	M 22 x 1.5	50 (37)
Side transmission cover to housing	M 8 x 35	23 (17)
Bearing cover to bearing assembly	M 6 x 15	
Drive pinion bearing assembly to housing	M 10 x 35	50 (37)
Plug for ATF ducts	M 14 x 1.5	25 (18)
Fastening nut to bearing assembly	M 36 x 1.5	250 (184)
Crown wheel to differential housing	M 10 x 1.25	85 (63), and Loctite 262

3 Technical Data

Torque specifications for transmission suspension

Location	Thread	Tightening torque Nm (ftlb.)
Transmission support to body	M 10 x 70	46 (34)
Transmission support to transmission (fastening nut)	M 12 x 1.5	85 (63)
Side member to transmission	M 12 x 1.5 x 65	85 (63)
Side member to transmission support (fastening nut)	M 10	30 (22)
Console to transmission	M 8 x 35	23 (17)

Torque specifications for ATF lines and cooler

Location	Thread	Tightening torque Nm (ftlb.)
Bracket to headlight holder	M 6	10 (7)
Console to wheel house	M 8	23 (17)
Tension strut to console	M 6	10 (7)
Bar to ATF cooler	M 6	10 (7)
ATF cooler to engine oil cooler	M 6	10 (7)
Horn holder to wheel house	M 6	10 (7)
ATF lines to transmission (union nut)	M 18	30 (22)
Joints on ATF lines (union nuts)	M 18	30 (22)

3 Technical data

Torque specifications for gear selecting system

Locaion	Thread	Tightening torque Nm (ftlb.)
Lock nut to clevis of selector lever cable	M 5	6 (4)
Selector lever mount to body	M 6 x 16	10 (7)
Holder for selector lever cable to transmission	M 8	23 (17)
Cable slide housing to switch plate (Keylock)	M 4	2.5 (2)
Keylock cable to ignition lock	M 10 x 1	2.5 (2)
Shiftlock to selector lever housing	M 5	6.5 (5)
Lift solenoid to Shiftlock housing	M 4	2.5 (2)

Removing and installing the torque converter

Removing

1. Remove transmission.

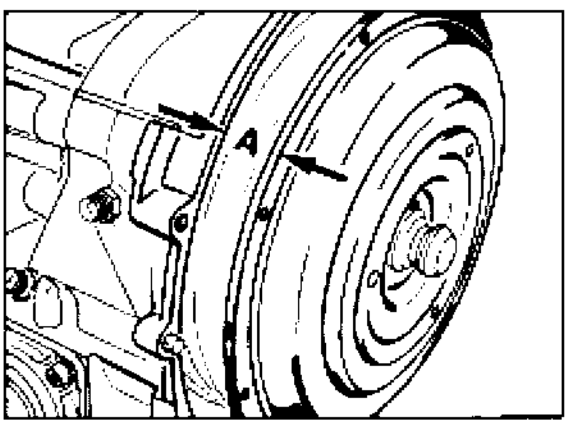
2. Remove converter, with transmission in horizontal position.

Note

Do not damage converter bearing assembly and rotary shaft seal.

Installing

1. Carefully insert converter, with transmission in horizontal position. Turn the converter to and fro until the gear toothing engages and the installation position is reached.



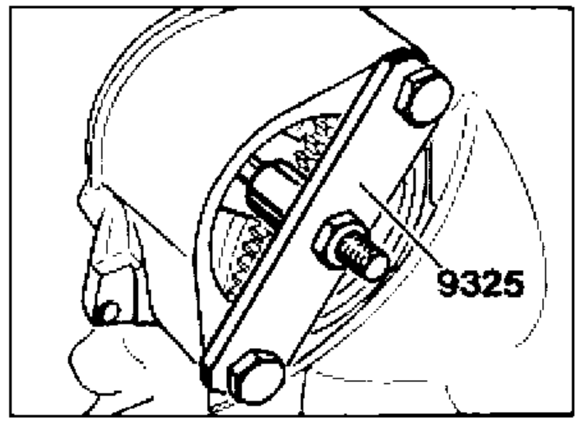
412-32

A = approx. 25 mm

Note

If the torque converter is not installed in the correct position, both the torque converter and the ATF pump may be damaged when the engine is connected to the transmission.

2. Secure converter against falling out with special tool **9325**



413-32

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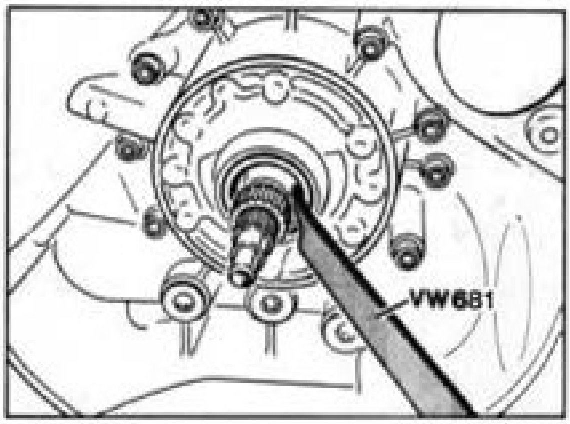
32 47 19 Removing and installing torque converter seal ring

Removing

1. Remove transmission and converter.

2. Lever out sealing ring with **VW 681**



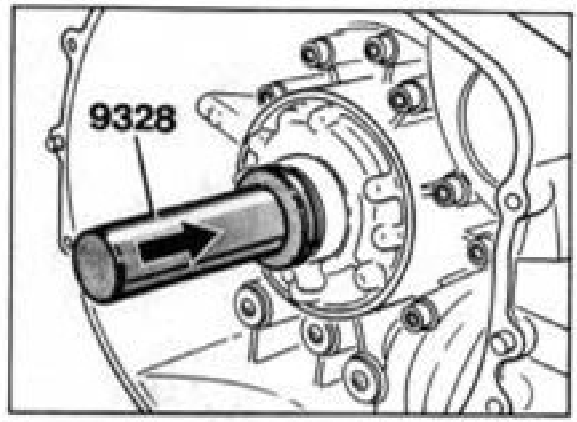
419-38

Installing

Installation takes place in reverse order.

1. Wet sealing lip with ATF.

2. Press in sealing ring with special tool **9328** as far as it will go.



420-38

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Test point	DTC	Title	Fault effect	Page
31	70	Fault in torque converter clutch	Torque converter clutch always open	37 - 85

Note

Diagnosis of the torque converter clutch is active as from the 1997 model.

Fault, fault code	Possible causes, elimination, notes

Test point 31

Torque converter fault

Fault code 70

Torque converter clutch always open

Fault possibility: mechanical/hydraulic fault in transmission

1) Check ATF level and correct if necessary (refer to 911 Carrera (993) Workshop Manual, Page 37 - 101).

2) Erase fault memory and perform a test drive.

The diagnostic test conditions are achieved if:
- the torque converter clutch is electrically closed
- engine speed < 3008 rpm
- engine torque > 200 Nm (148 ftlb)
- ATF-temperature between 40° C and 95° C

3) Read out fault memory again.
The following areas have to be checked if the fault is still present:
- ATF supply
- torque converter
- hydraulic control unit (jammed spool valves)
- transmission (slipping clutches)

Printed in Germany - 20, 1997

Checking the ATF fluid level

The prescribed fluid level is extremely important for perfect functioning of the automatic transmission.

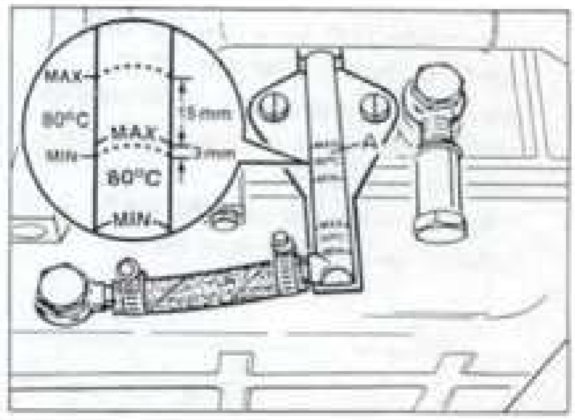
Preconditions for checking:

- Transmission underbody cladding removed
- Vehicle must be horizontal
- Engine operating at idling speed
- Hand brake applied
- Selector lever in position "P"
- ATF temperature 80°C

Check ATF level at 80°C.

Note

The ATF capacity was increased by 0.5 l. The ATF level therefore rises and the 80°C check marks move further up on the oil level tube. Refer to imaginary and dotted lines in the close-up insert.
When checking the ATF level, make sure the fluid level is between those two lines (maximum level **15 mm above** the **80°C MAX mark** present, lowest level approx. **3 mm below** the **80°C MAX mark** present).



1958-38

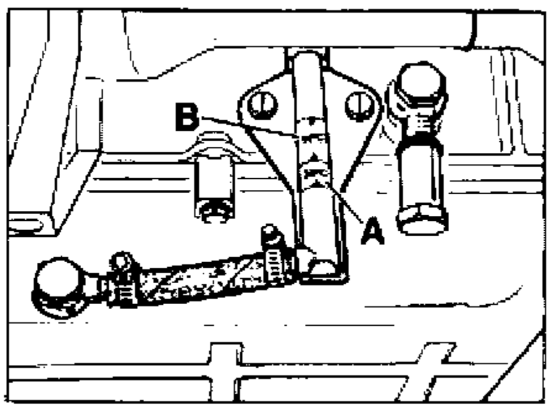
A = invalid 80°C mark

The exact ATF temeprature can be determined with the system tester 9288.

If necessary, top up missing ATF fluid via the quick-fill device (also refer to page 37-103).

Modifications from model year '95

Since model year '95, a new oil level tube with **50°C** markings has been fitted.
The ATF fluid level should be checked using the same procedure as before but at an ATF temperature of **50°C**.
The liquid level must be within the 50°C indication range (see B in illustration).



2036-38

A = indication range for 30°C ATF temperature
B = indication range for 50°C ATF temperature

37 02 55 Changing ATF fluid

Capacity: approx. 9.5 l
Change quantity: approx. 3.5

Oil type:
ATF-Dexron IID

The ATF fluid must be changed and the ATF strainer cleaned every 40,000 km

When changing the ATF fluid, the vehicle must be horizontal and the engine switched off.

Drain ATF fluid, remove ATF pan and ATF strainer (refer to page 38-113).

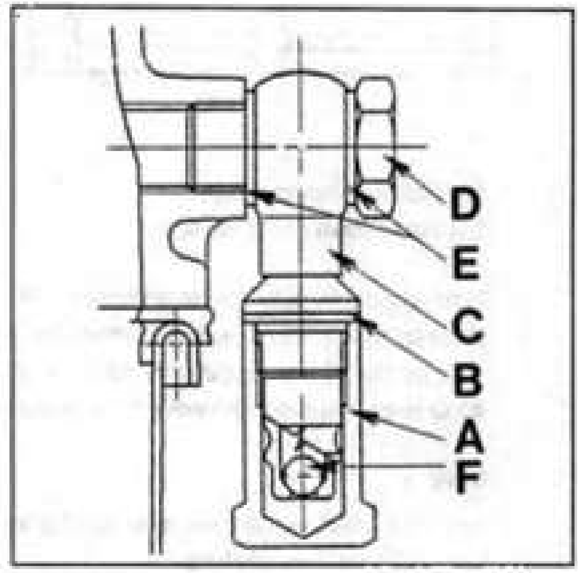
Thoroughly clean the strainer and pan.

Fit the ATF strainer with a new O-ring. Tighten the fixing screws with **8 Nm (6 ftlb)**.

Fit the ATF pan with seal. Tighten the fixing screws with **8 Nm (6 ftlb)**.

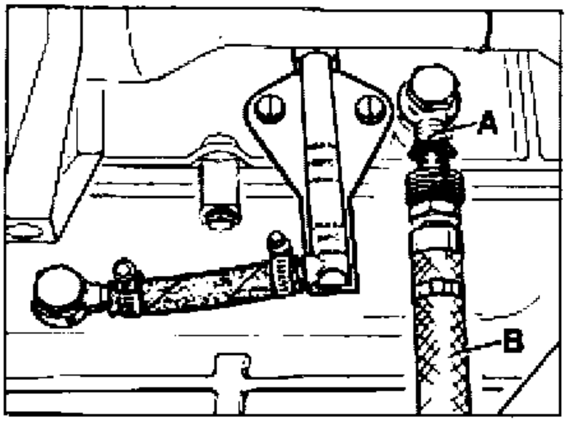
Fill with ATF fluid:

First, fill ATF fluid up to the 30°C max. mark via the quick-fill connection with the engine stationary.



391-38

A = Hexagon cap nut (tightening torque 30 Nm = 22 ftlb)
B = Sealing ring (replace)
C = Quick-fill connection



388-38

A = Quick-fill connection
B = Hose from filling device

Start engine in selector lever position "P" and allow to run at idle speed. Observe the ATF level in the oil level pipe and top up to the 30°C max. marking immediately if required.

Note

The ATF level in the transmission changes with the fluid temperature.

Drive the transmission warm and check the ATF fluid at 80° C (from mod. '95 at 50° C). **See note on page 37 - 101.**

The exact ATF temperature can be determined with the system tester 9288.

Printed in Germany - 12, 1995

37 31 19 Removing and installing multifunctional switch on transmission

Removal

1. Set selector lever to position "N".

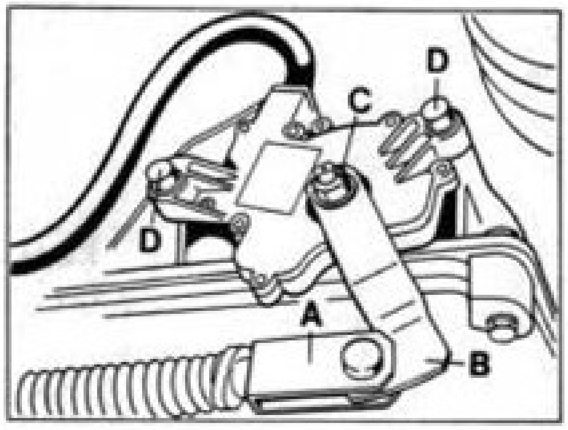
2. Remove transmission undertray.

3. Remove rear underside panel.

4. Remove left rear hot air pipe.

5. Disconnect selector lever cable from actuating lever.

6. Remove actuating lever.

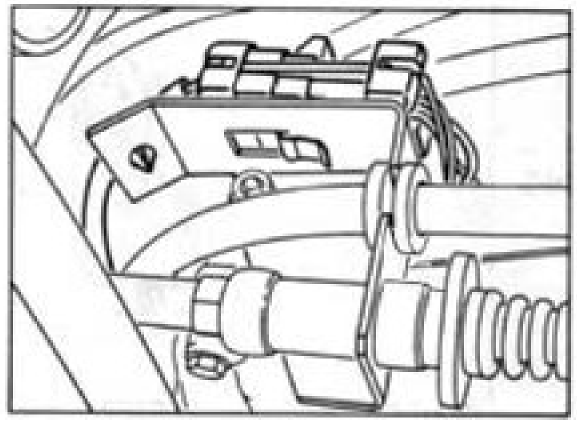


381-37a

A = selector lever cable
B = actuating lever
C = hexagon nut (M 8 x 1) with washer
D = fastening screws

7. Press retaining lugs for wire retainer "A" together and lift holder out upwards.

8. Detach, unlock and disconnect connector.



1881-37

A = wire retainer

9. Unscrew fastening screws for multifunctional switch completely and remove switch.

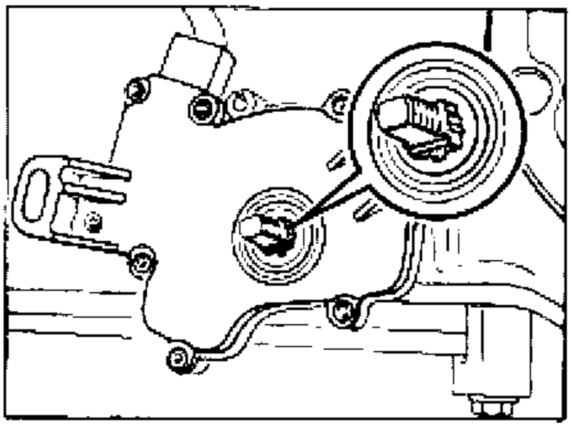
Installation

For installation, proceed in reverse order.

Tightening torques:

Multi-functional switch to transmission = 10 Nm
Actuating lever to selector shaft = 15 Nm

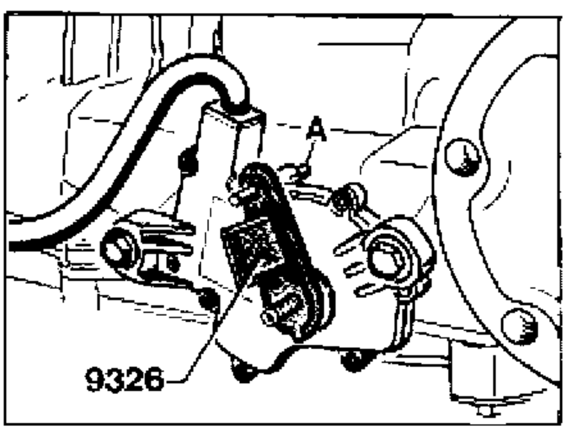
1. Set selector shaft to position "N" (turn shaft anti-clockwise up to the stop and then back two clicks) and place switch in the correct position.



1884-37

2. Adjusting multifunctional switch.

   Push pointer of special tool **9326** onto the selector shaft and turn the switch until the locating pin can be pushed into the fixing hole of the switch. Tighten the mounting screws to **10 Nm** (7 ftlb.) in this position.



A = locating pin

2. Check, and if necessary adjust, setting of selector lever cable.

37 15 15 Adjsuting cable for selector device

1. Move selector lever to position "P".

2. Set multi-function switch to position "P". To do so, press actuator lever of switch back up to stop.

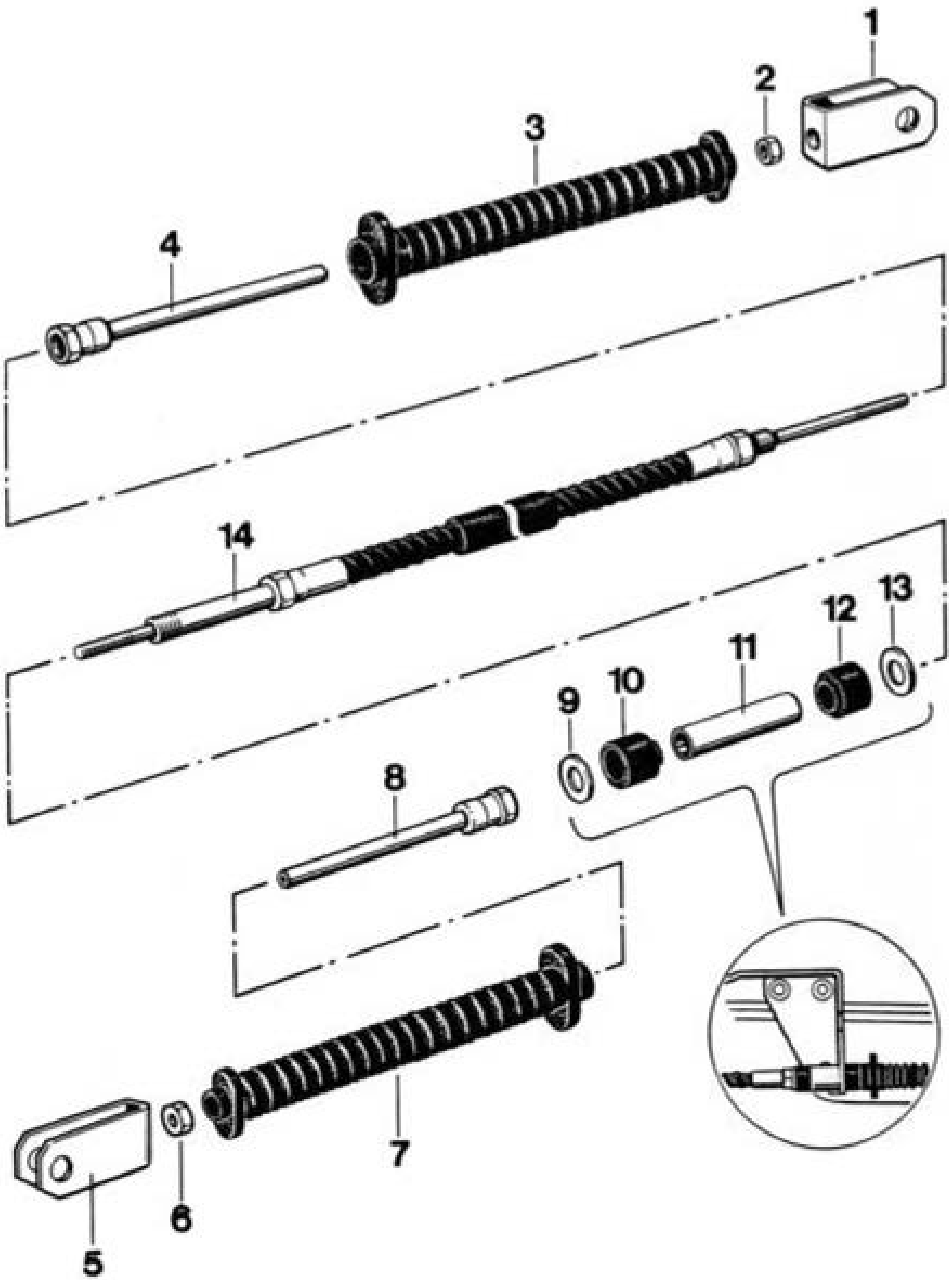
3. Set cable length at clevis so that the bolt can be installed free from stress.

4. Check adjustment by shifting through all the gears and confirming that gear is displayed on speedometer. In addition change gate from "D" to "M". This must be possible with one shmooth, straightline movement.

5. Mount snap ring for bolt on actuator lever.

Printed in Germany - 12, 1995

37 15 19 Removing installing cable for selector mechanism



No.	Designation	Qty.	Note: Removal	Installation
1	Fork head	1		Screw on tol half length of thread on cable
2	Hexagon nut	1		Tighten to 5 Nm (4.4 ftlb)
3	Gaiter seal	1		
4	Guide tube	1		
5	Fork head	1		
6	Hexagon nut	1		Tighten to 6 Nm (4.4 ftlb)
7	Gaiter seal	1		
8	Guide tube	1		
9	Washer	1		
10	Rubber mount	1		Install in correct position
11	Spacer tube	1		
12	Rubber mount	1		Install in correct position
13	Washer	1		
14	Cable	1		Readjusrt

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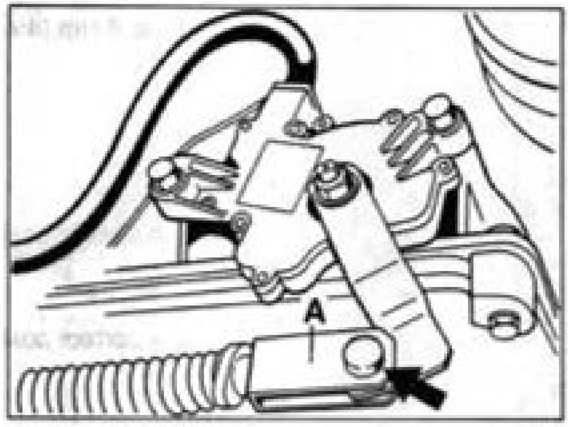
Instructions for removal and installation

Removal

1. Remove transmission undertray and underside panel.

2. Disconnect cable on operating lever and remove fork head.

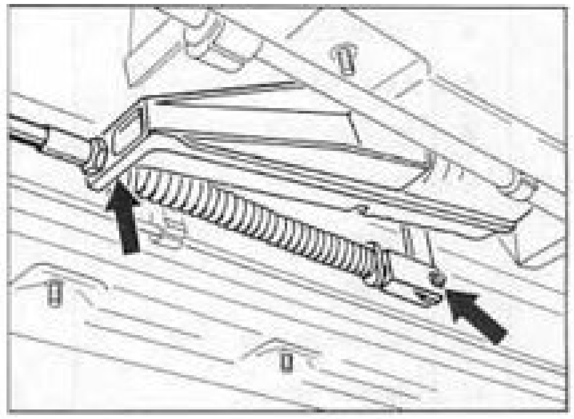


381-37

A = Fork head

3. Disconnect guide tube on bracket.

4. Disconnect cable on selector lever casing and disconnect guide tube.



1066-37

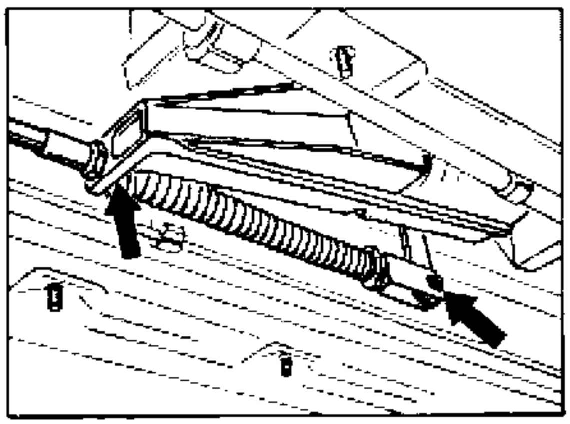
Installation

1. Installation is carried out in reserve order.

2. Adjust calbe for selector mechanism (see page 37 - 107).

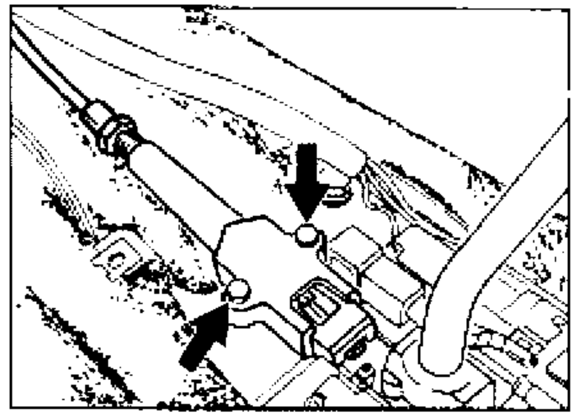
37 10 19 Removing and installing gear selecting system

1. Disconnect battery.
2. Remove center underside panel
3. Disengage selector lever cable from deflection lever and undo guide tube.



1066-37

4. Unscrew release button and pull off selector knob.
5. Remove center console.
6. Disconnect cable valve body from switch plate (only for keylock models).



1067-37

7. Disconnect connectors for switch plate.
8. Unscrew four mounting screws (M 6) and take out selector lever operator from above.

Installation

Tigtening torques:

Gear selecting system to body = 10 Nm (7 ftlb)

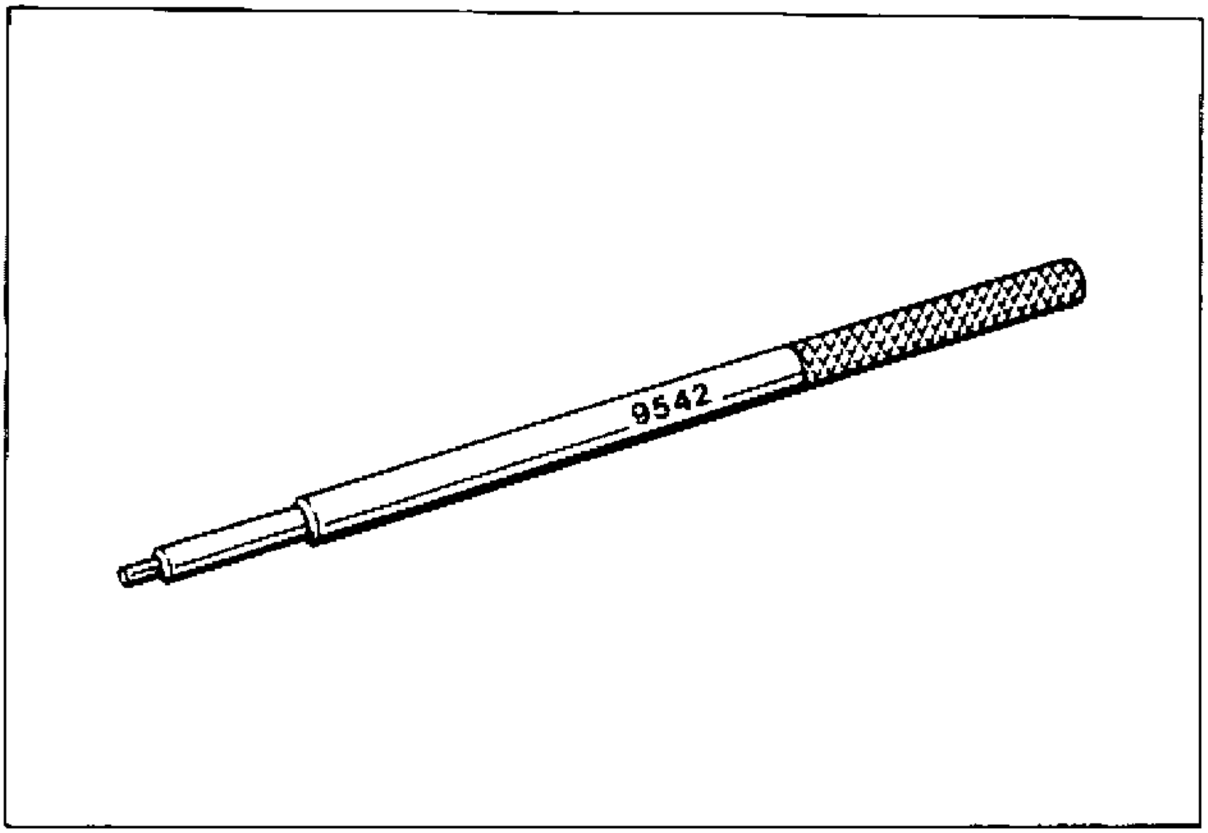
Cable valve body to switch plate = 2.5 Nm (2 ftlb)

1. Install in reverse order.
2. Press selector knob manually until is up against the stop, making sure the twist lock engages in the cutout in the selector lever.
3. Check release button for smooth operation. It must return into the home position by itself.
4. Check selector lever cable adjustment and readjust if required.
5. Check operation of keylock and shiftlock

Printed in Germany – V, 1994

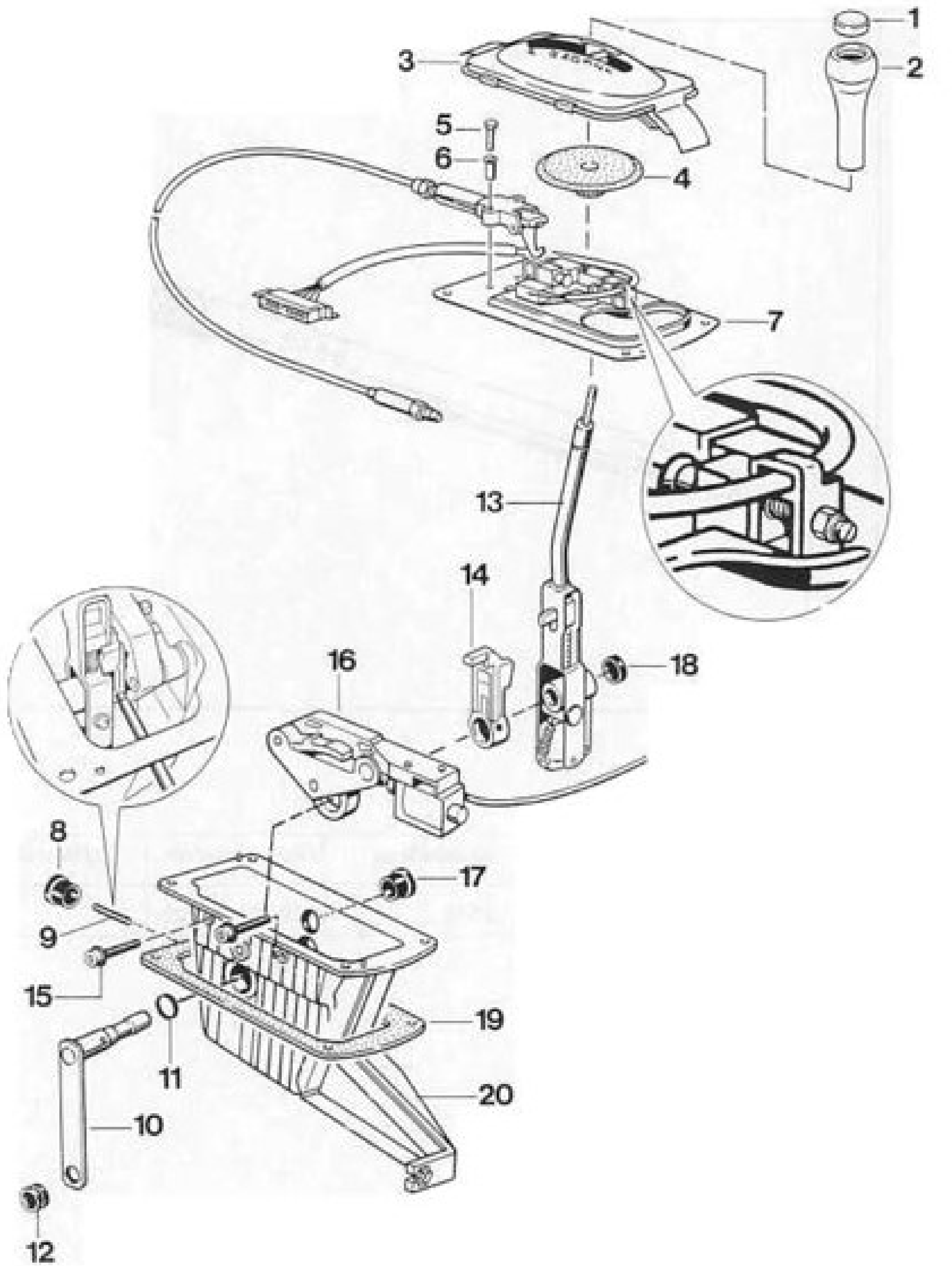
37 10 37 Dismantling and assembling gear selecting system

Tools



No.	Designation	Special tool	Order number	Explanation
	Assembly mandrel	9542	000.721.954.20	

37 10 37 Dismantling and assembling gear selecting system



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No.	Designation	Qty.	Note: Removal	Installation
1	Release button	1	Screw off manually	Must return to home position automatically after it has been actuated
2	Selector knob	1	Pull off manually	Push up to stop in correct position. Twist lock must engage in cutout in selector lever
3	Shroud	1	Can only be removed from below with the center console removed	
4	Cover	1		
5	Hexagon head screw	2		Tighten to 2.5 Nm (2 ftlb)
6	Sleeve	2		
7	Switch plate	1		
8	Plug	1		
9	Tensioning sleeve	1	Drive out using Special Tool 9542	Bore of driver and relay shaft must match
10	Deflection lever	1		
11	Seal	1		
12	Rubber mount	1		
13	Selector lever	1	Shift into manual speed selection gate	
14	Driver	1		
15	Hex socket head bolt	2		Tighten to 6.5 Nm
16	Shift-Lock	1		
17	Plug	1		
18	Cover	1		
19	Gasket	1		
20	Housing	1		

Note

Coat all sliding surfaces with low-temperature grease (e.g. Shell S 6508)

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Notes on assembly:

Note

Change from "D" to "M" and from "M" to "D". Stright-line movement without catching must be possible. In addition, there may be up to 0.4 mm play on the selector lever in the + and – direction with "M" selected. Adjustments can be made using the selector lever reset switch (see exploded drawing, item 7).

37 87 19 Removing and installing solenoid for shiftlock

Note

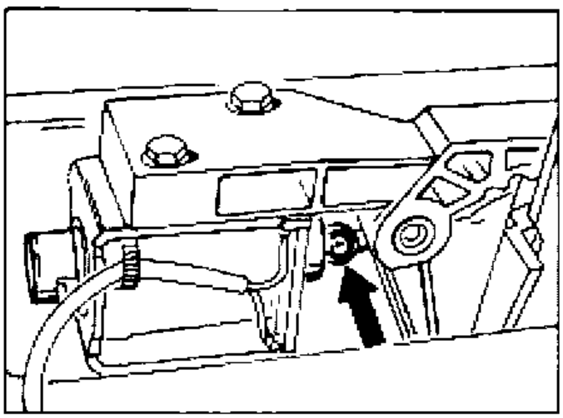
When checking the solenoid electrically, be sure to observe correct polarity
Terminal 1 = Positive
Terminal 2 = Negative

Removal

1. Remove center console and gear selecting system.

2. Remove switch plate.

3. Using a suitable tool, press connecting rod carefully off the solenoid.



4. Screw out mounting screws and take off solenoid.

Installation

1. Set selector lever to "1" position.
2. Grease ball and ball socket with low-temperature grease.

3. Clip connecting rod to solenoid, retaining the connecting rod in correct position using a suitable wire hook or marking tool and pushing the free lift solenoid carefully into the ball socket.

4. Using sleeves and hexagon head bolts, fit solenoid to gate in such a manner that it remains free to slide in an axial direction.

5. Adjusting the solenoid:

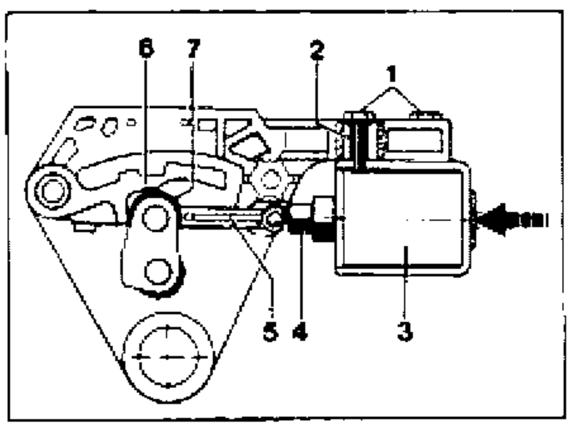
   Set selector lever to position „P".

   Slide lift solenoid all the way back in the slots of the gate.

   Push iron core from the solenoid towards the pawl until it contacts the stop and locate it in this position.

   Slide the actuated solenoid axially until the idler contacts the stop of the pawl.

   Tighten hexagon head bolt to 2.5 Nm (2 ftlb) in this position.



1 - Hexagon head bolt
2 - Sleeve
3 - Solenoid
4 - Iron core
5 - Connecting rod
6 - Pawl
7 - Idler

6. Check operation of shiftlock.

Printed in Germany – V, 1994

37 13 19 Removing and installing keylock bowden cable

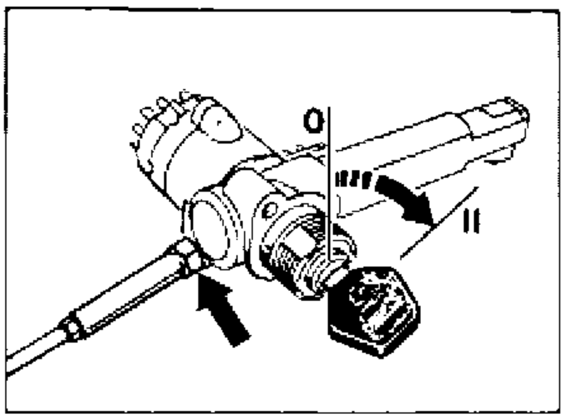
Removal

1. Disconnect battery.
2. Remove complete center console, knee guard and side nozzle.
3. Undo Central Information System and leave it supended on the wiring.
4. Turn ignition lock to position „2" (Ignition on).

Note

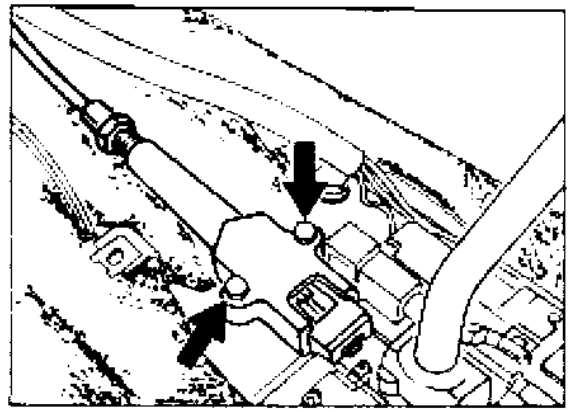
The ignition lock must be in position „2". If it is left in any other position, both the lock and the bowden cable may be damaged.

5. Unbolt bowden cable from ignition lock.



1068-37

6. Undo mounting screws of valve body.



1067-37

7. Disengage bowden cable from pedal floorboard and from bracket for center console.

Installation

1. Assemble in reverse order, observing the following points.
2. Turn ignition lock to position „2" (Ignition on) and fit bowden cable to ignition lock.
   Tightening torque: 2.5 Nm (2 ftlb)
3. Fit bowden cable valve body using sleeves and hexagon head bolts.
   Tightening torque: 2.5 Nm (2 ftlb)
4. Set selector lever to position „P" and turn ignition lock to position „0".

Note

If the ignition lock cannot be turned to the 0 position, the bowden cable must be readjusted.

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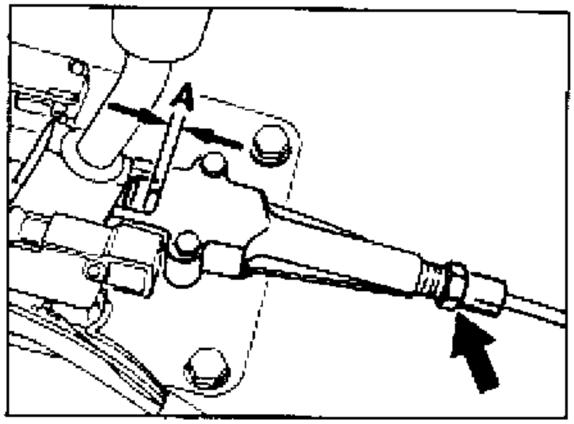
5. Adjust bowden cable, turning the cable sleeve on the valve body until the lock slide reaches a setting of 2 + 0.5 mm (refer to Fig.).



1081-37

A = 2 + 0.5 mm

6. Check operation of keylock and shiftlock.

37 13 01 Checking keylock and shiftlock

Checking the keylock (ignition key lock)

Selector lever position	Position of release button	Keylock operation	Ignition key
P	not actuated	no	rotary, may be pulled off
P	actuated	yes	cannot be turned to pulloff position
R-N-D-3-2-1	not actuated actuated	yes	cannot be turned to pulloff position

Checking the keylock (selector lever lock)

Ignition key position	Release button status	Keylock operation	Selector lever
Pulled off or pulled position	locked	yes	locked
Position 1 or 2	not locked	no	not locked

Checking the shiftlock

Selector lever position	Ignition	Shiftlock operation	Brakes	Selector lever
P-R-N-D-3-2-1	off	no	actuated not actuated	not locked
P and N	on	yes	not actuated	locked
P and N	on	yes	actuated	not locked

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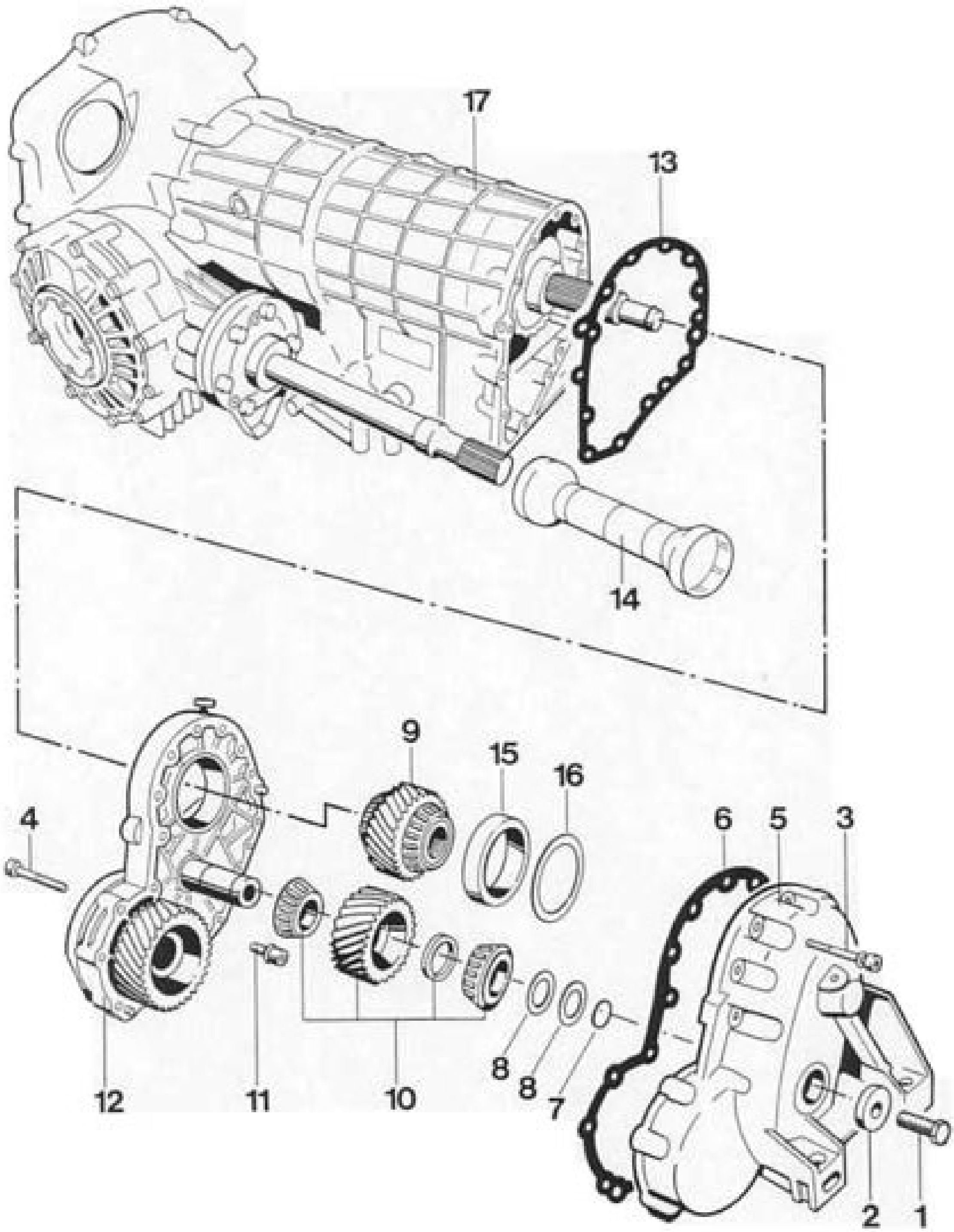
37 48 19 Removing and installing the intermediate plate



No.	Designation	Qty.	Note: Removal	Installation
1	Hexagon screws	1		Tighten with 46 Nm
2	Washer	1		
3	Fillister head screw	11		Tighten with 23 Nm
4	Fillister head screw	6		Tighten with 23 Nm
5	Front transmission cover	1		
6	Seal	1		Replace
7	O-ring	1		Replace, oil with ATF
8	Shim	X	Note thickness for re-installation	
9	Gear wheel with tapered-roller bearings	1		
10	Gear wheel with bearing assembly	1		Set by the manufacturer.
11	Fillister head screw	2		Tighten with 23 Nm
12	Intermediate plate	1		
13	Seal	1		Replace
14	Protective tube	1		Large diameter to final drive
15	Taper roller bearing outer race	1	Remove with internal puller (e.g. Schrem 60 - 70)	Heat transmission cover to approx. 120 deg. C and press in to stop
16	Adjusting shim	X	Record thickness for reinstallation	
17	Automatic transmission with final drive	1		

Note

The reduction gear is only available as a complete set (intermediate plate, transmission cover and adjusted spur gears). It is not necessary to adjust the tension on the tapered roller bearing.

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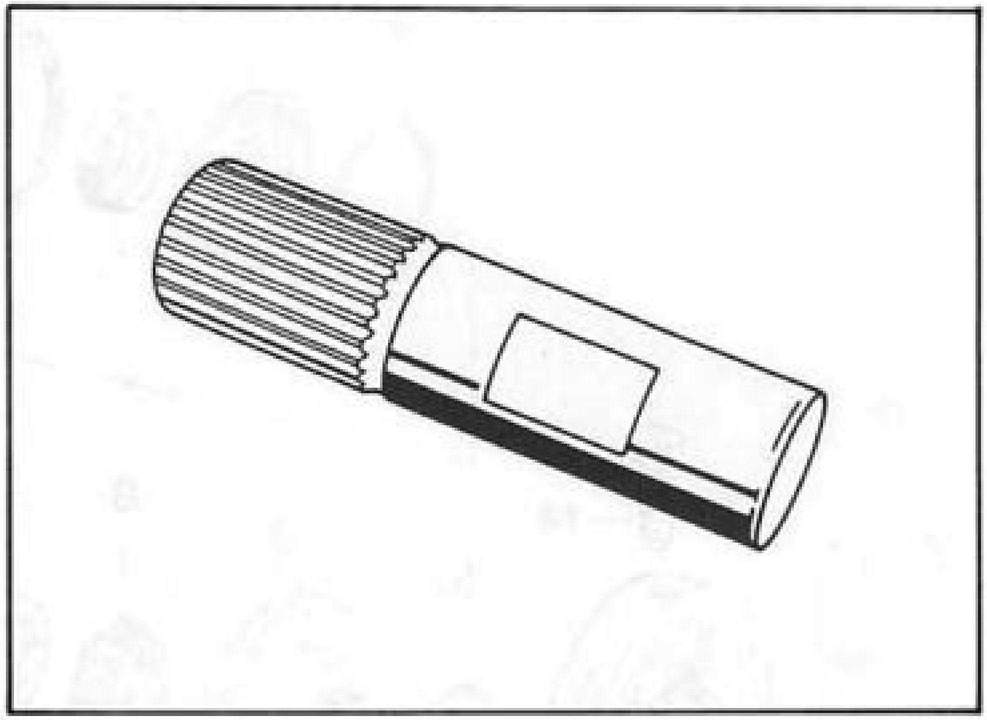
37 48 37 Dismantling and assembling intermediate plate

Tools



No.	Designation	Special tool	Order number	Explanation
	Retaining device	9340	000.721.934.00	

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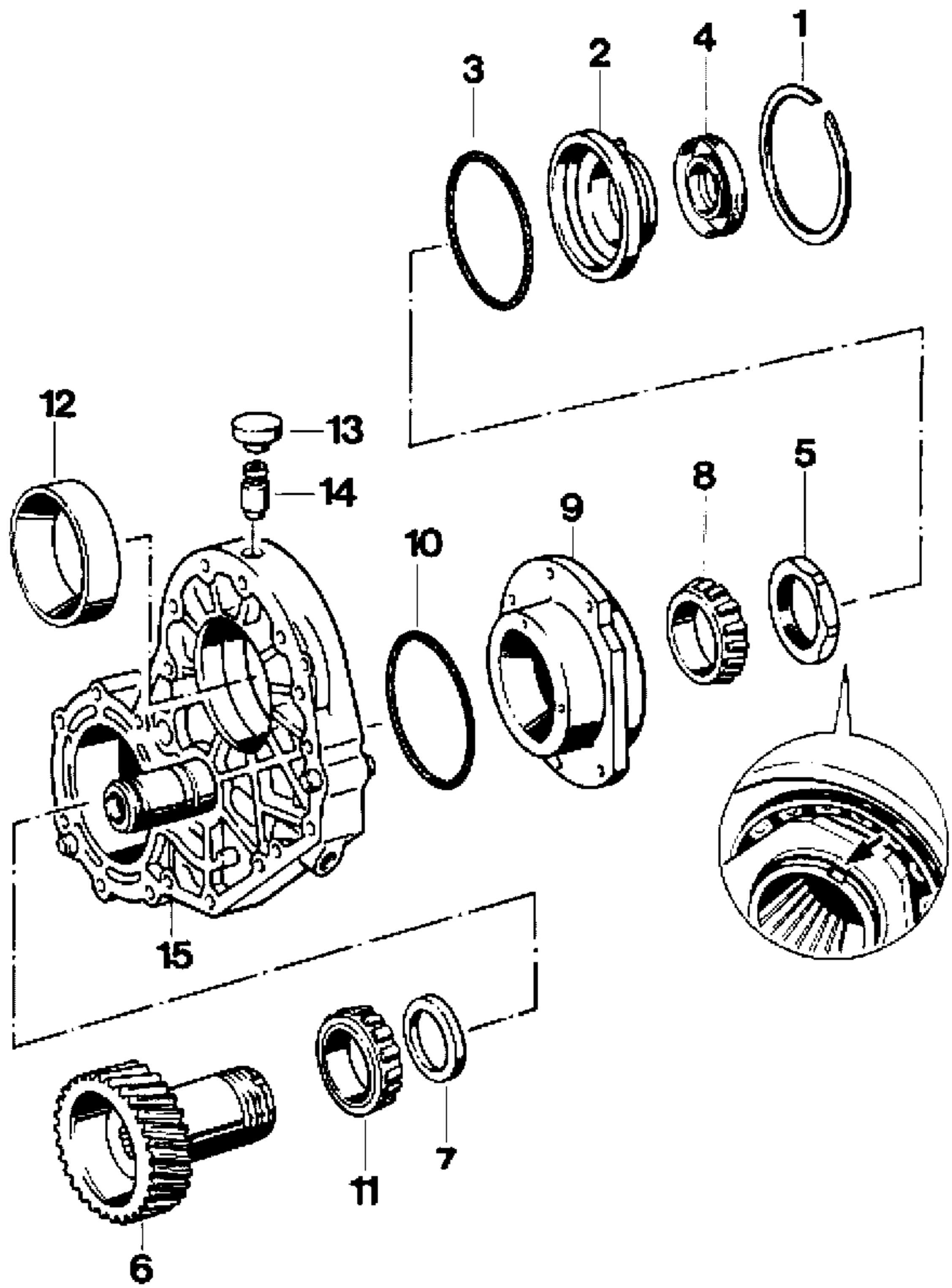
34 48 i7 D ma g and assembling intermediate plate



No.	Designation	Qty.	Note: Removal	Installation
1	Snap ring	1		
2	Bearing cover	1	Grab at lugs to lift out	
3	Round seal	1		Replace, coat with ATF fluid
4	Shaft seal	1		Fitting depth 2.0 ± 0.5 mm
5	Lock nut	1	Fit Special Tool 9340 into vise. Put intermediate plate into position and undo nut.	Tighten to 250 Nm (184 ftlb). Lock by upsetting the flange (2 x 180°)
6	Helical gear	1	Press out using a hydraulic press.	Replace only as a set (along with intermediate gear and drive gear)
7	Adjuster ring*	X	Record thickness for reassembly	Thickness can only be determined by manufacturer
8	Inner race of taper roller bearing*	1	Mark for reassembly	Heat to approx. 120°C and press on
9	Bearing cover *	1		
10	Round seal	1		Replace, coat with ATF fluid
11	Inner race of taper roller bearing*	1	Remove across assembly bore of helical gear. Mark for reassembly.	Heat to approx. 120°C and press into place
12	Outer race of taper roller bearing	1		Heat intermediate plate to approx. 120°C and press into place
13	Breather cover	1	Lever off	
14	Breather tube	1	Pull out	Press in to stop
15	Intermediate plate	1		

Note

The parts identified by an * have been preadjusted by the manufacturer.

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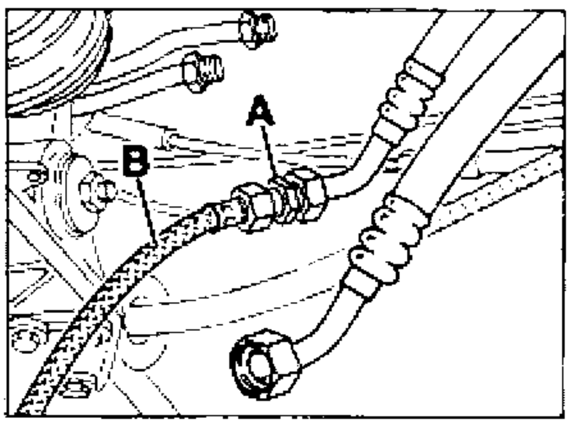
38 60 29 Flushing ATF cooler and lines

Information

If the ATF is carbonized, or if there is sludge or evidence of lining abrasion in the ATF pump, it is not sufficient merely to repair or replace the gearbox, the ATF cooler and line system must be flushed with ATF.

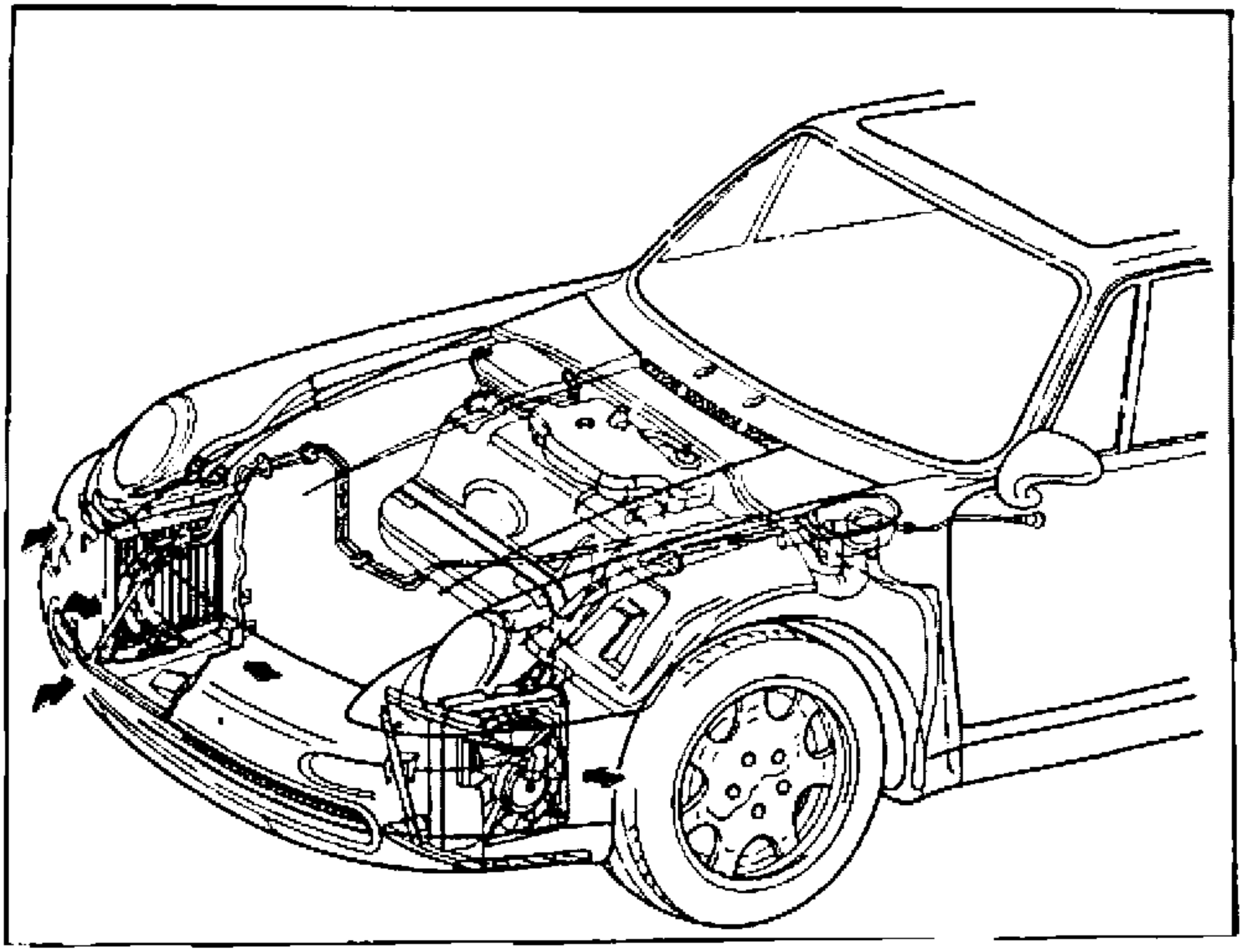
Attach additional hose of ATF filling device (see Workshop Manual Group 3/4) using special tool **9355/1** to ATF line with conventional twin connecting piece and flush out cooler and line system using filling device. Fluid must not be split on the ground.



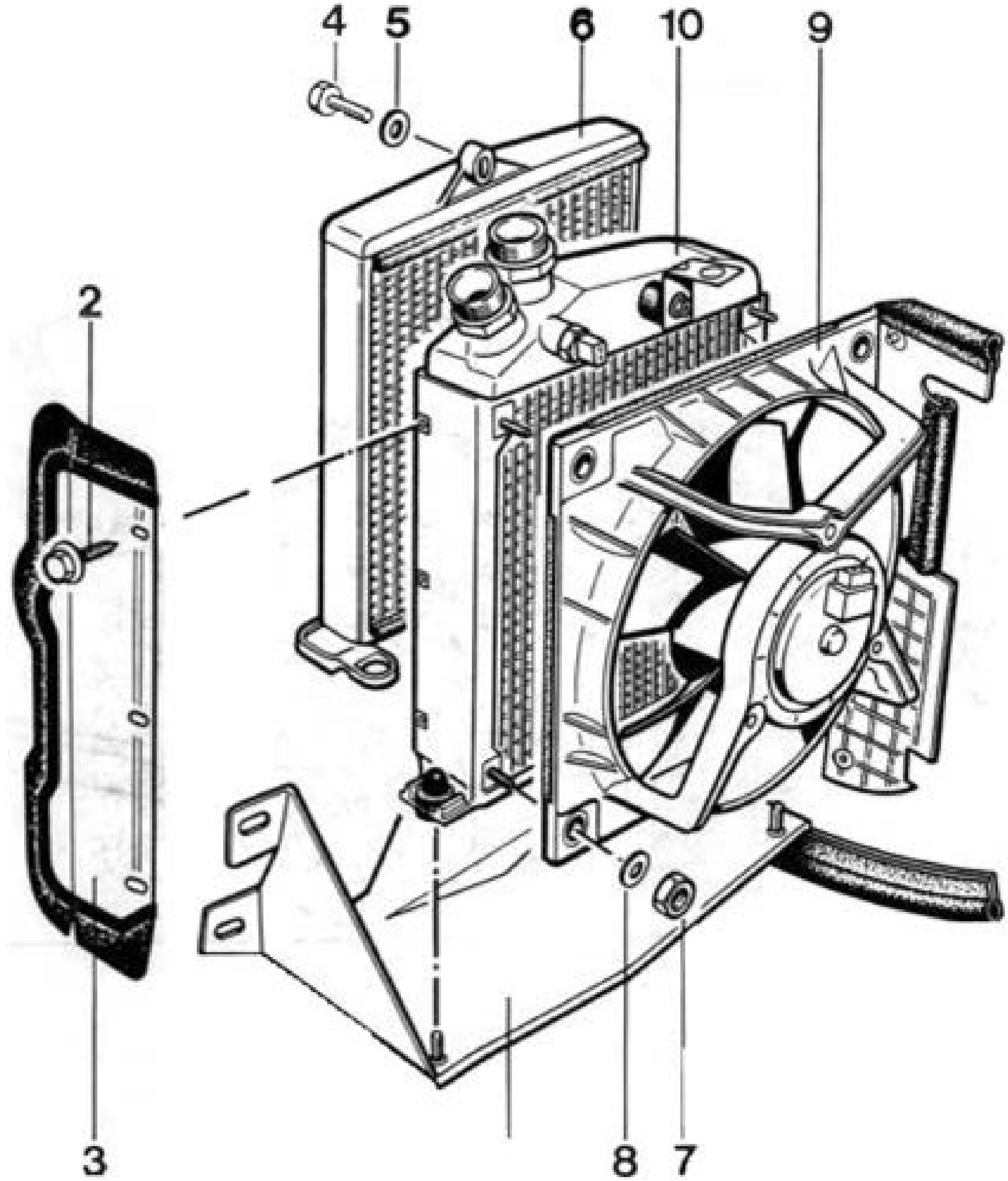
1883-37

A = Special tool 93551
B = Additional hose

38 60 R moving nd insta ng ATF ooler



38 60 R m ving and sta ling ATF coole



No.	Designation	Qty.	Note: Removal	Installation
1	Bracket	1		
2	Washer-and-screw assembly	3		
3	Air deflector	1		
4	Hexagon screw	1		Tighten to 10 Nm (7.3 ftlb)
5	Washer	1		
6	ATF cooler	1		

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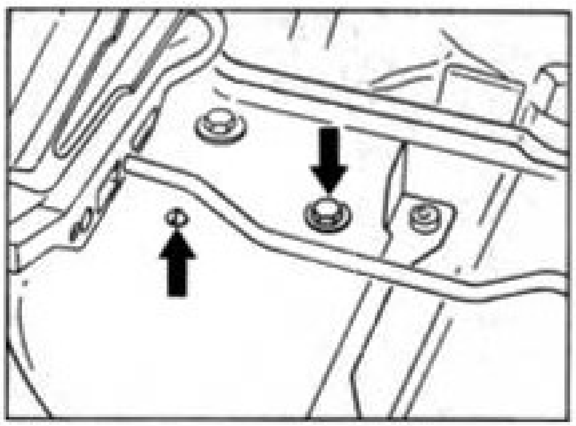
38 60 19 Removing and installing ATF cooler

Notes

The ATF cooler is installed in the wheel house in front of the right front wheel. It is mounted on the engine oil cooler and cooled by the two-stage electric fan of the engine oil cooler.

Removal

1. Remove right headlight, unclip wire retainer and unscrew hexagon head screw for cooler bracket.

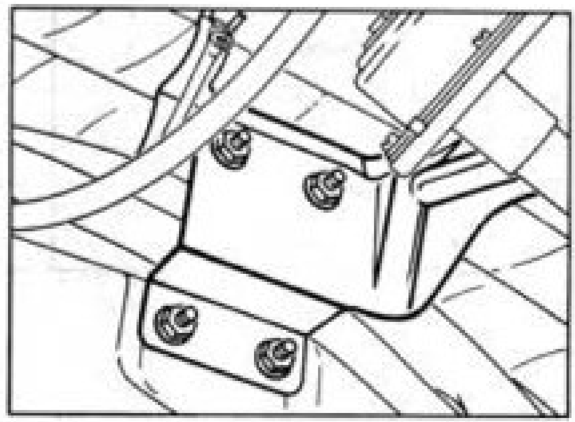


1877-38

2. Remove front wheel housing liner and lower part of front spoiler.

3. Pull connector off electric fan and remove indicator mount by turning it to the left (bayonet lock).

4. Unscrew horn mount from wheel house.

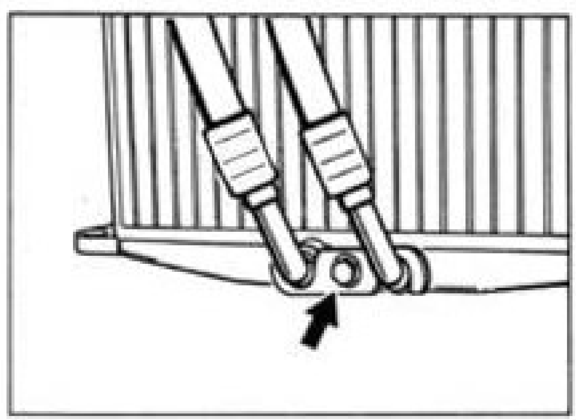


1878-38

5. Remove console. To do so, unscrew three hexagon nuts (M 8) from wheel house and one (M 6) for tension strut.

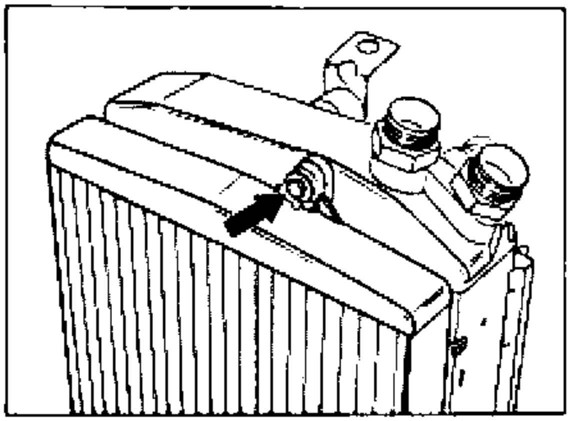
6. Remove air baffle.

7. Disconnect ATF lines from cooler.



1879-38

8. Unscrew hexagon head screw and remove ATF cooler (illustration shows cooler removed).



487-38

Note

Do not disconnect oil lines for engine oil cooler. The cooler remains in the wheel house.

Installation

Tightening torques, Nm (ftlb.

Bracket to headlight holder	= 10 (7)
Console to wheel house	= 23 (17)
Tension strut to console	= 10 (7)
ATF cooler to engine oil cooler	= 10 (7)
Bar to ATF cooler	= 10 (7)
Horn holder to wheel house	= 10 (7)

1. Adopt the reverse procedure for installation.

2. Replace O rings for ATF lines on cooler.

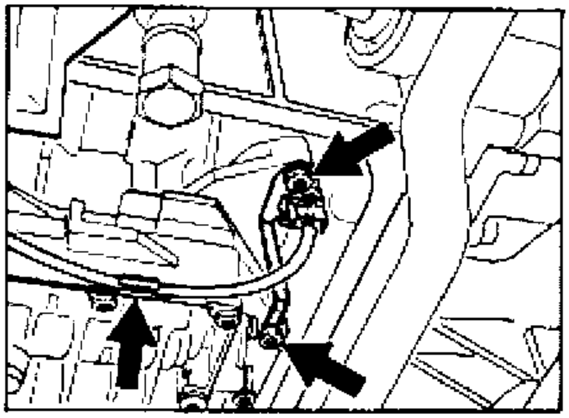
3. Check that the rubber edging of the fan housing and the console is correctly positioned.

4. Check, and if necessary top up, ATF level. (see page 37 - 101).

38 17 19 Removing and installing the inductive speed sensor

Removing

1. Remove the transmission underbody cladding and ATF pan (refer to page 38 - 113).
2. Remove the holder for the inductive sensor and pull out sensor



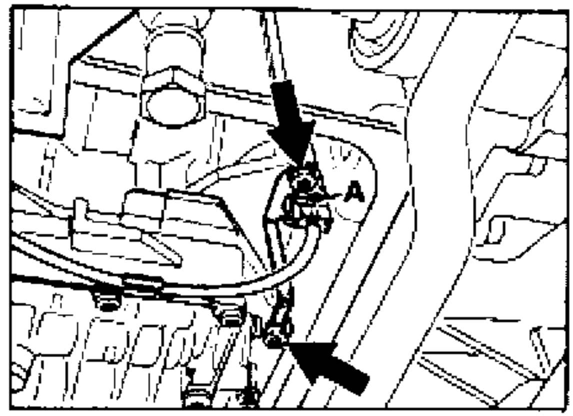
390-38a

Installing

Tightening torque:
Holder to control unit = 8 Nm

Installation takes place in reverse order.

Insert the sensor and mount the holding plate so that the lugs engage in the groove on the connector.



386-38b

A = Connector groove

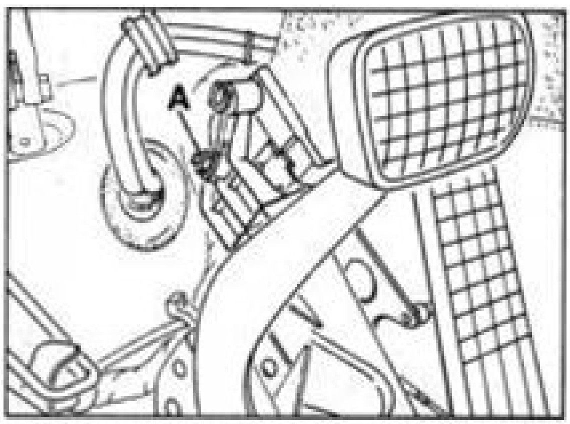
38 90 19 Removing and installing kickdown switch

Removal

The kickdown switch is installed on a console in front of the accelerator pedal.

1. Remove floor board.

2. Remove kickdown switch with mount, unplug connector and remove switch (pressing the retaining lugs).



2244-38

A = self-locking nut

Installation

1. Install the parts removed in reverse order.

2. Use new self-locking nut for mount.

3. Place kickdown switch in installation position. With the mounting nut loosened, the mount with switch must be pushed until the actuator arm touches the console when the switch is operated.

4. Tightening torque: mount to console: M 6 = 10 Nm (7.5 ftlb)

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38 90 15 Adjusting kickdown switch

Prerequisites:

The idle position of the accelerator cable must be correctly adjusted.

Kickdown switch in installation position. (see page 38 - point 3).

1. Connect system tester 9288.
2. Unhook pushrod from accelerator pedal.
3. Switch on system tester and select throttle valve in DME /actual values menu.
4. With the pushrod unhooked, push the accelerator pedal down to the stop (throttle valve open up to stop) and read **throttle position 1** off on system tester.

Note

With the throttle valve open (accelerator pedal fully depressed), the position is 84° ± 30.

5. Hook pushrod back onto accelerator pedal.
6. Push gas pedal down as far as possible **without** operating kickdown switch and read off **throttle position 2** on system tester.
   The position indicated must be 3° less than position 1 (tolerance ± 10).

Note

If the angle is too large, set the pushrod to a shorter position.
If the angle is too small, set the pushrod to a longer position.

After resetting the length of the pushrod, check the angle. It must not be smaller than 79°.

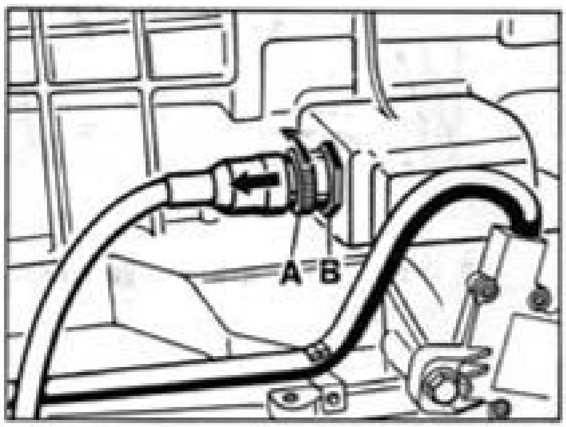
38 18 19 Remov. and installing the wiring harness for the transmission

Removing

1. Remove the transmission underbody cladding.

2. Remove multi-functional switch (see page 37-105).

3. Disconnect the connector from the transmission socket. To do this, turn the bayonet lock to the left and disconnect the connector:



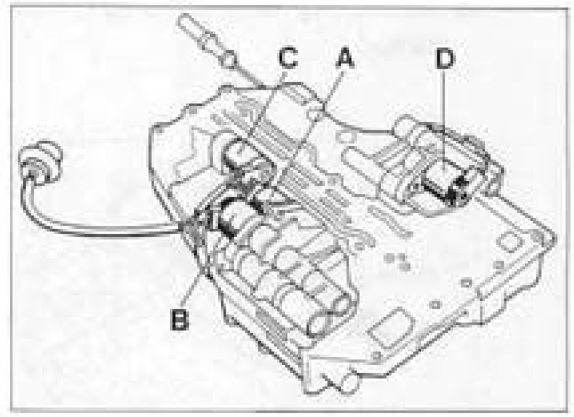
387-38a

A = Bayonet lock
B = Hexagon nut (a/f 30)

4. Unscrew the hexagon nut for the transmission socket with a suitable extension.

5. Remove hydraulic control unit with wire harness (see also page 38-113).

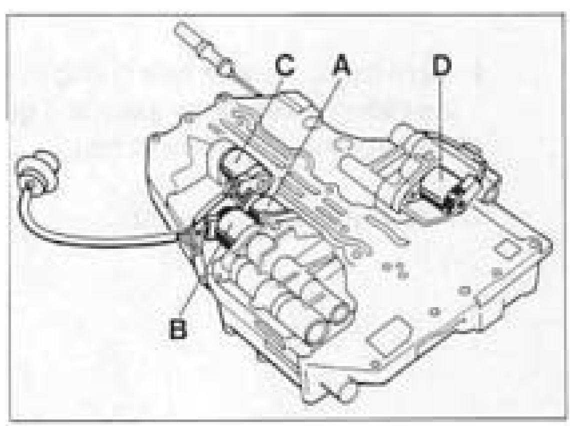
6. Mark the push-on sleeves for re-installation and pull off from the solenoid valves.



383-38

Installing

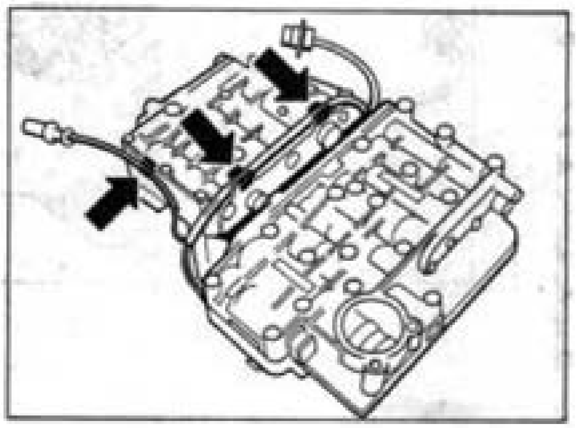
1. Push the push-on sleeves for the solenoid valves up to the stop. Pay attention to cable colors.



383-38

	Solenoid valves	Cable colors
A	Solenoid valve 1	(grey-violet)
B	Solenoid valve 2	(green-violet)
C	Solenoid valve 3	(red-violet)
D	Pressure regulator	(blue-violet)

2. Route wiring harness and hang in cable clamps.



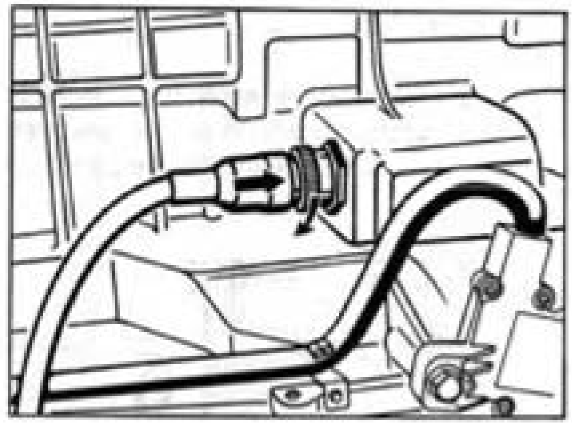
384-38

3. Place hydraulic control unit on a suitable surface (e.g. transmission jack) at installation height.

4. Insert the socket with **new O-ring** so that the flattened side faces upwards. Tighten hexagon nut with **12 Nm (9 ftlb)**.

5. Install hydraulic control unit.

6. Connect wiring harness with transmission socket. To do this, carefully insert the connector in the socket (fits only in one position) and lock by turning the bayonet lock to the right.



387-38b

7. Install the multi-functional switch and readjust.

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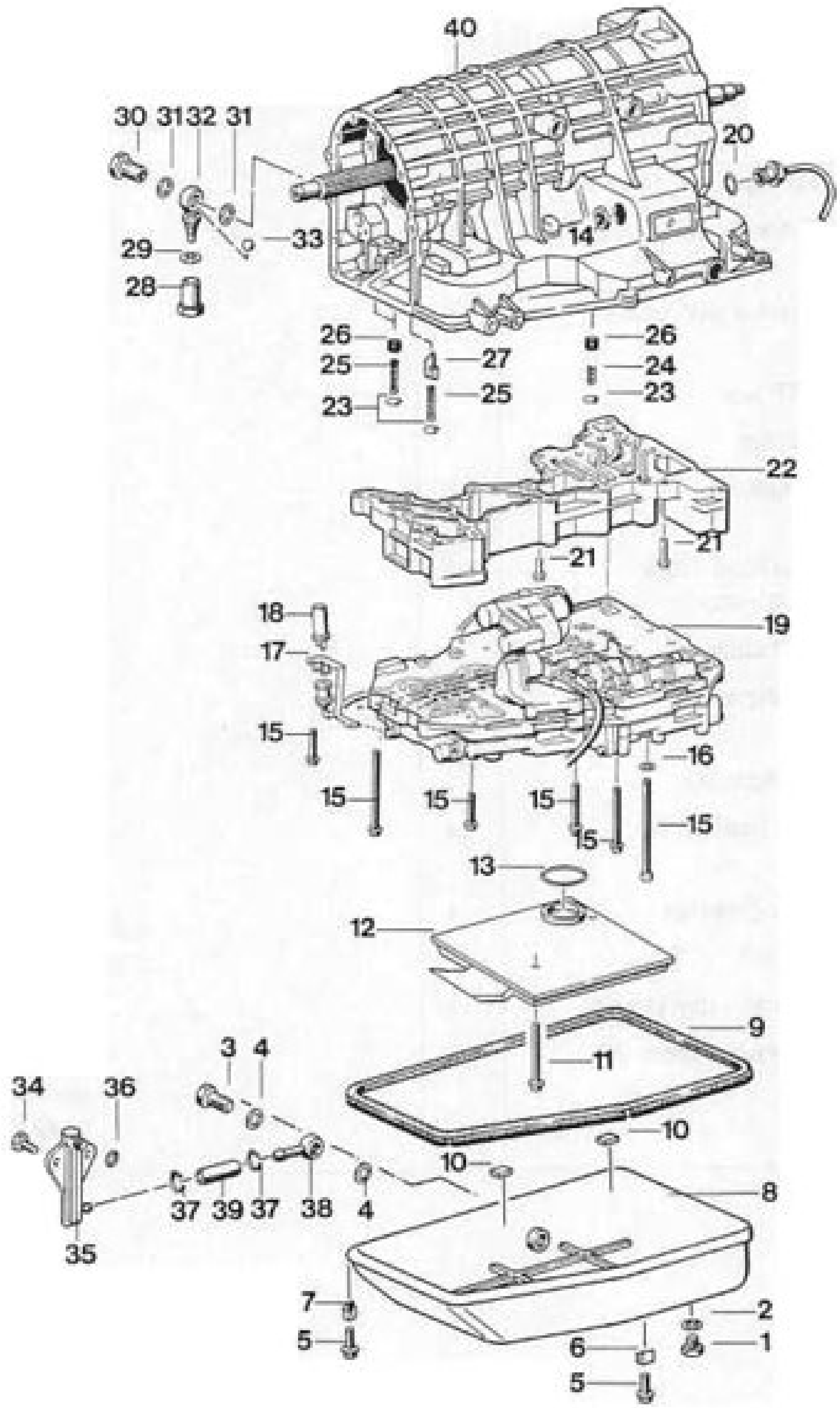
38 77 19 Dismantling and assembling hydraulic control unit



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No	Designation	Qty.	Note: Removal	Installation
1	Plug	1		Tighten to 40 Nm (29 ftlb)
2	Seal	1		Replace
3	Banjo bolt	1		Tighten to 40 Nm (29 ftlb)
4	Seal	2		Replace
5	Hexagon head bolt	6		Tighten to 6 Nm (4 ftlb)
6	Bracket (straight leg)	2		Short legs must force on the ATF pan
7	Bracket (curved leg)	4		Short legs must force on the ATF pan
8	ATF pan	1		
9	Gasket	1		
10	Magnet	2		Place into grooves in ATF pan
11	Pan head screw (M 6 x 65)	3		Tighten to 8 Nm (6 ftlb)
12	ATF strainer	1		
13	Round seal	1		Replace. Make sure that position is correct
14	Hexagon nut	1		Tighten to 12 Nm (9 ftlb)
15	Pan head screw	14		Observe correct length, tighten to 8 Nm (6 ftlb)
16	Spring washer	1		
17	Bracket	1		
18	Inductive rpm pickup	1		
19	Hydraulic control unit	1		The straight surface on the harness socket must point towards the outside

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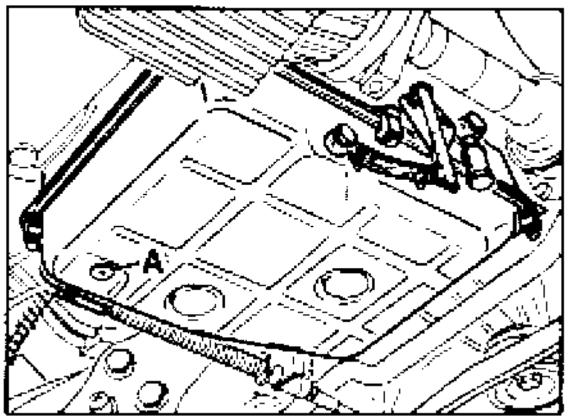
38 77 19 Removing and installing the hydraulic control unit

Removing

The transmission wiring harness remains on the transmission.

1. Remove the transmission underbody cladding.

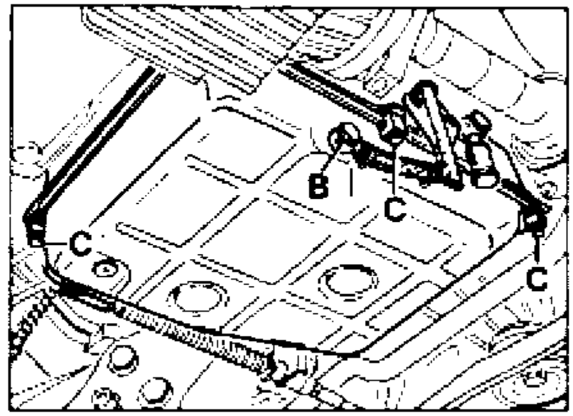
2. Drain off ATF fluid.



386-38a

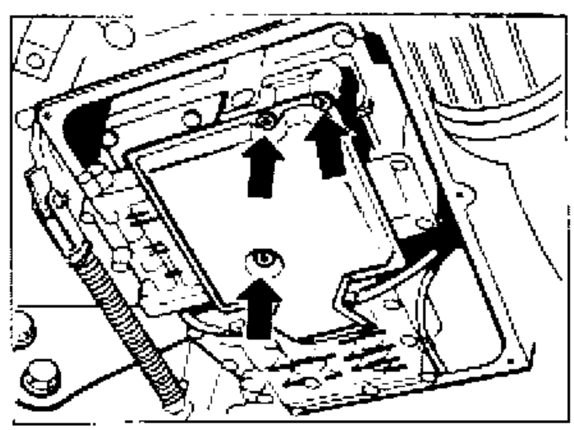
A = Drain screw with sealing ring

3. Remove ATF pan. To do this, unscrew hollow screw for oil level pipe and six fixing screws.



B = Hollow screw
C = Fixing screws

4. Remove ATF strainer with **Torx insert T 27.**



382-38

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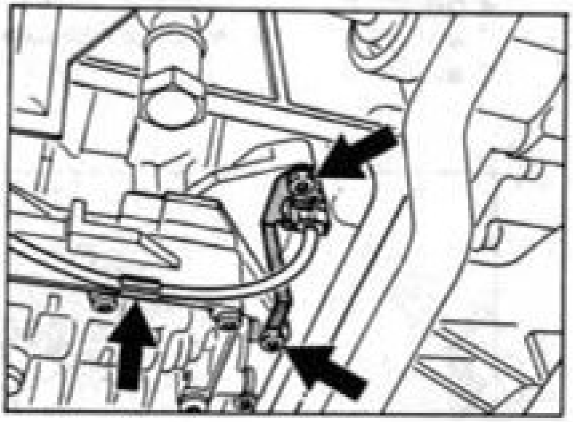
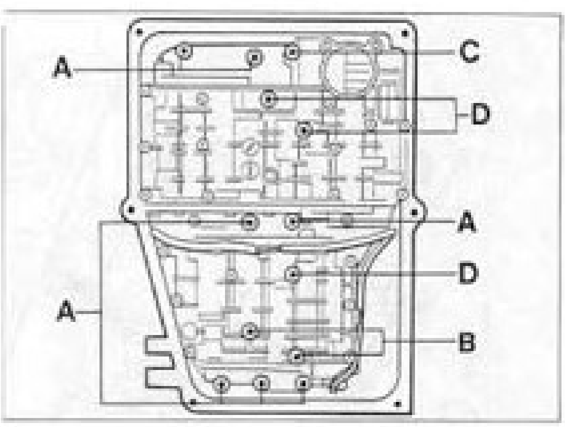
5. Remove inductive speed sensor and pull wiring harness out of holding clamps.



390-38a

6. Unscrew 13 fixing screws (head diameter 12 mm) **with Torx insert T 27** and lower hydraulic control unit only so far as to guarantee that the wiring harness is not subjected to any tension.
   Place the hydraulic control unit on a suitable surface (e.g. transmission jack).



399-38

Note

The hydraulic control unit must not be suspended from the wire harness.

7. Mark the push-on sleeves for re-installation and pull off from the solenoid valves.

8. Pull the wiring harness out of the holding clamps and remove the control unit.

Installing

Installation takes place in reverse order.

1. Tightening torques:

Control unit to transmission	= 8 Nm (6 ftlb)
ATF strainer to control unit	= 8 Nm (6 ftlb)
ATF pan to transmission	= 6 Nm (4 ftlb)
Drain screw to ATF pan	=40 Nm (29 ftlb)
Hollow screw to ATF pa	=40 Nm (29 ftlb)

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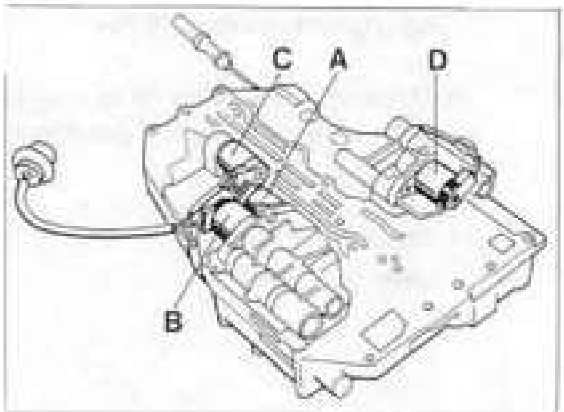
2. Push on push-on sleeves for solenoid valves up to the stop. Pay attention to cable colors.

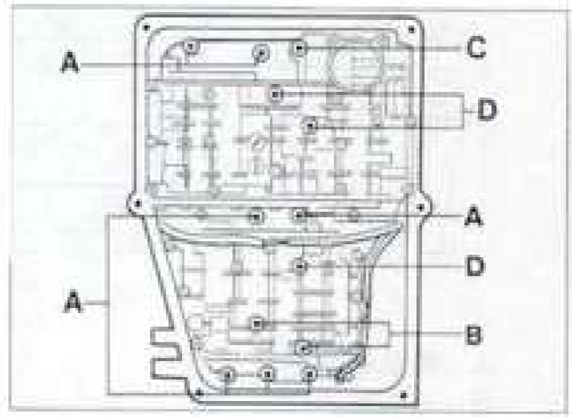


383-38

	Solenoid valves	Cable colors
A	Solenoid valve 1	(grey-violet)
B	Solenoid valve 2	(green-violet)
C	Solenoid valve 3	(red-violet)
D	Solenoid valve (pressure controler)	(blue-violet)

3. Mount the hydraulic control unit so that the pin of the notched disk projects into the recess of the selector slide.

4. Screw in the fixing screws for the hydraulic-control unit and counter slightly. Pay attention to screw lengths.



399-38

A = Screw length 80 mm
B = Screw length 65 mm
C = Screw length 115 mm
D = Screw length 60 mm

5. Position hydraulic control unit. To do this, move notched disk to position 1 (1st gear) and push hydraulic control unit back until it rests against the notched disk. Tighten fixing screws with **8 Nm** in this position.

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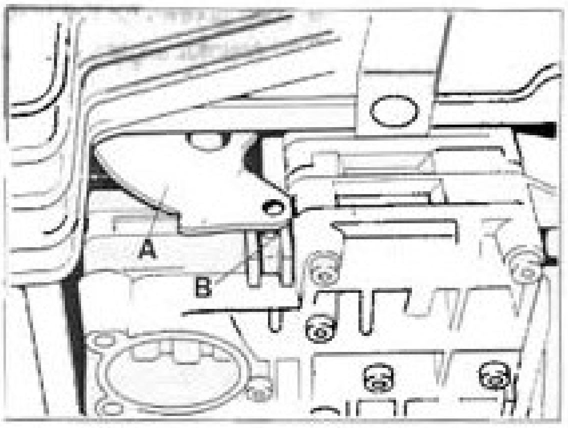
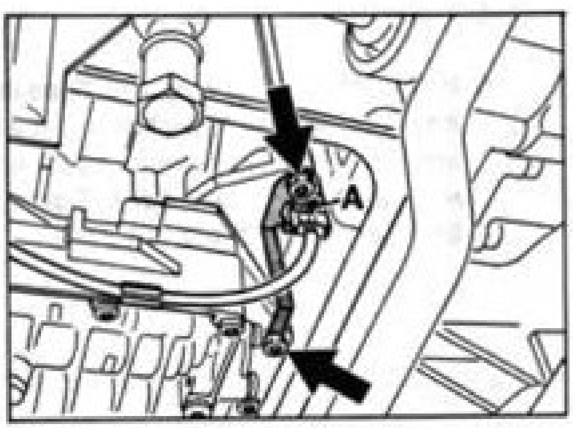


385-38

A = Notched disk
B = Selector slide

6. Insert pulse sensor and mount holding plate so that the lugs engage in the connector grooves.



390-38b

7. Fit ATF strainer with O-ring. Tightening torque **8 Nm**.

8. Place two magnets in the beads of the ATF pan, fit seal and secure the pan with the holding brackets so that the short legs press onto the ATF pan.

Note

The two holding brackets with the straight legs must be mounted at the side.

9. Screw in ATF drain screw with new sealing ring. Tightening torque **40 Nm**.

10. Screw in hollow screw for oil level pipe with new sealing rings. Tightening torque **40 Nm**.

11. Fill with ATF fluid (refer to page 38-103).

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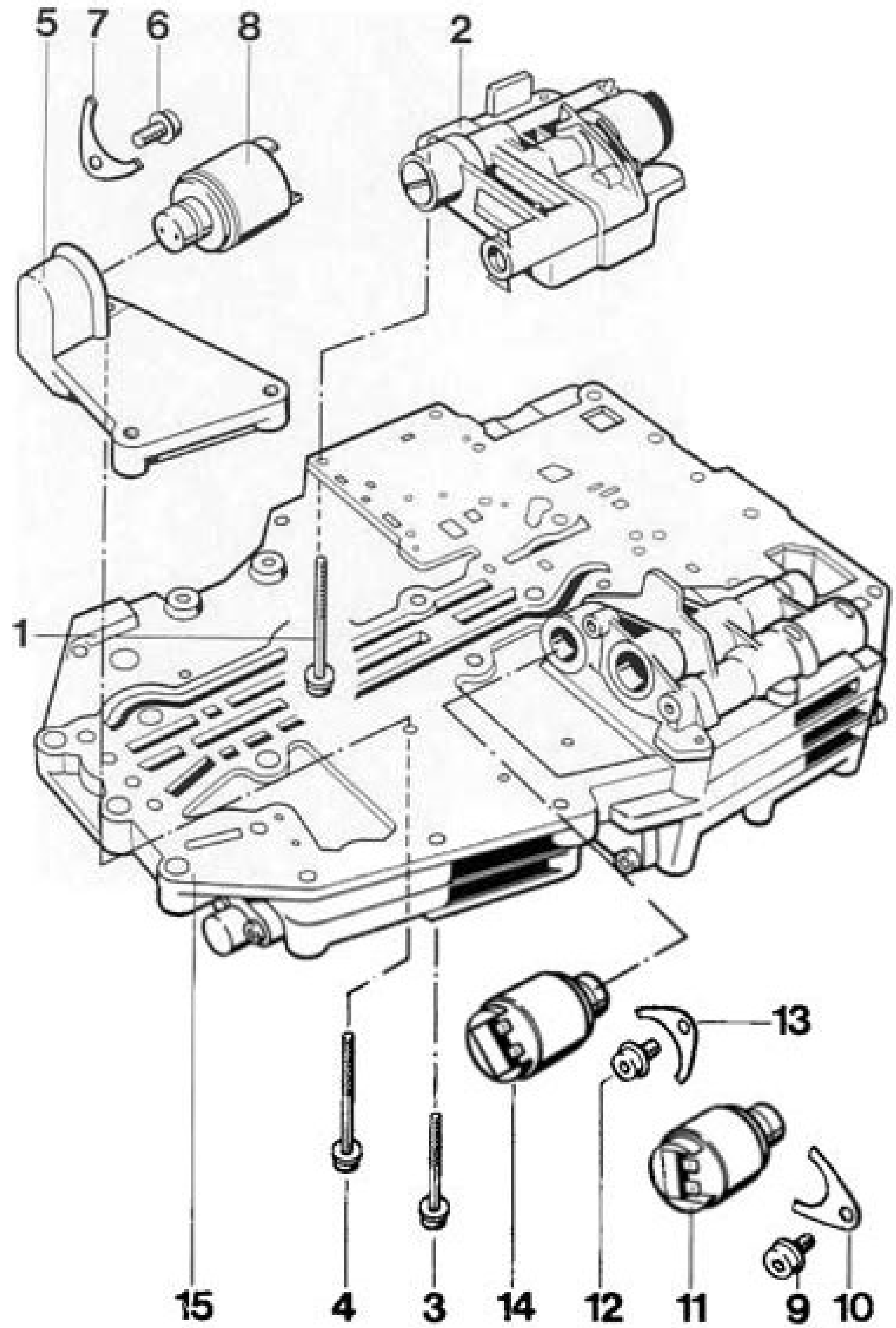
38 89 20 Removing and installing solenoid valves



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No.	Designation	Qty.	Note: Removal	Installation
1	Fillister head screw	5	Remove with Torx insert T 27	Tighten with 5 Nm (4 ftlb)
2	Housing with pressure regulator	1		Set by the manufacturer, only replace complete
3	Fillister head screw	2	Remove with Torx insert T 27	Tighten with 5 Nm
4	Fillister head screw	1	Remove with Torx insert T 27	Tighten with 5 Nm
5	Holder	1		
6	Fillister head screw	1	Remove with Torx insert T 27	Tighten with 5 Nm
7	Holding plate	1		Fit in correct position, lugs point to holder No. 5
8	Solenoid valve 3	1	Mark for re-installation	Do not confuse with solenoid valves Nos. 11 and 14
9	Fillister head screw	1	Remove with Torx insert T 27	Tighten with 5 Nm
10	Holding plate	1		Fit in correct position, lugs point to housing
11	Solenoid valve 2	1	Mark for re-installation	Do not confuse w[illegible] solenoid valves Nos. 14 and 8
12	Fillister head screw	1		Tighten with 5 Nm
13	Holding plate			Fit in correct position, lugs point to housing
14	Solenoid valve 1	1	Mark for re-installation	Do not confuse with solenoid valves Nos. 11 and 8
15	Hydraulic control unit	1		

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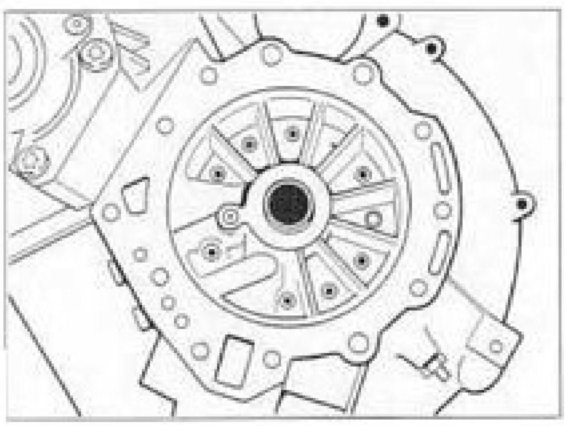
38 56 19 Removing and installing the ATF pump

Removing

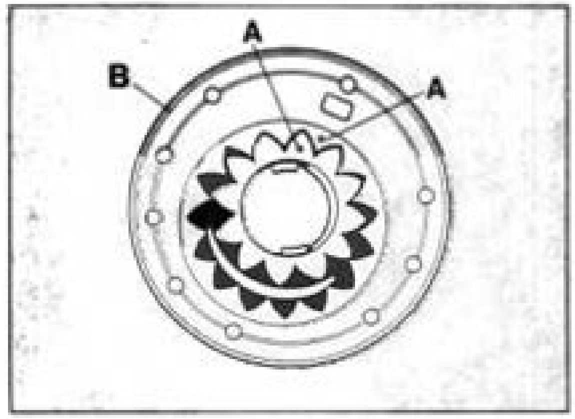
1. Remove transmission.

2. Remove torque converter and final drive.

3. Unscrew fixing screws with **Torx insert T27** and remove pump. To do this, screw in two screws opposite each other and carefully drive out the pump by gentle blows on the screw heads



406-38



405-38

A = Installation markings
B = O-ring

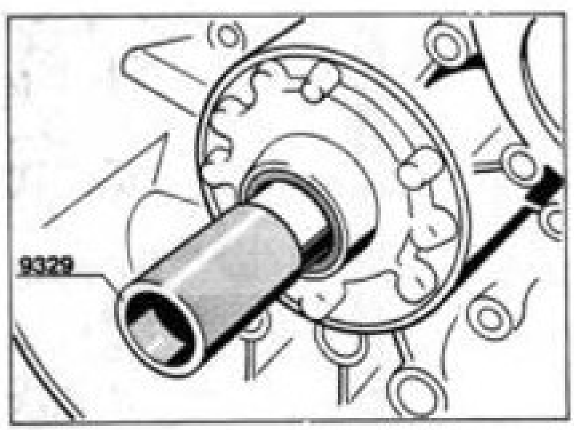
Installing

1. Check bearing bush for run-in scores or damage. The pump must be replaced if there are signs of damage.

2. Oil both pump wheels with ATF fluid and place in the housing so that both installation markings are visibly facing up.

3. Install pump with new O-ring and centering sleeve. Tighten fixing screws with **10 Nm**.

4. Check pump for easy operation. It must be possible to turn the pump by hand with special tool **9329** without it catching.



421-38

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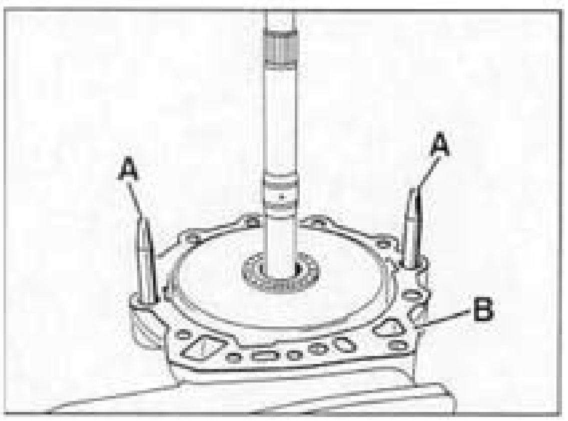
38 10 05 Determining the end clearance

Specification: 0.2...0.4 mm

Note

Check the end clearance whenever the transmission has been reassembled and readjust if required, using adjusting washer No. 12 (refer to page 39 - 111).

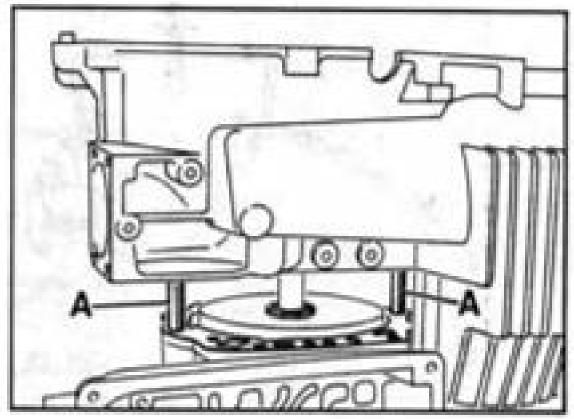
1. Assembly transmission (complete with spur gear drive and front transmission cover).
2. Screw centering pins **9321** into transmission case and apply some grease (vaseline) to stick paper gasket to sealing surface.



964-38

A - Centering pins **9321**
B - Gasket

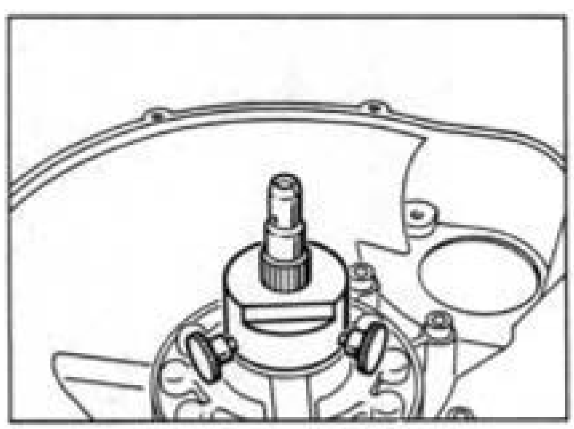
3. Using some grease (use vaseline), stick the removed adjusting washer to the final drive and put housing carefully into position.



961-38

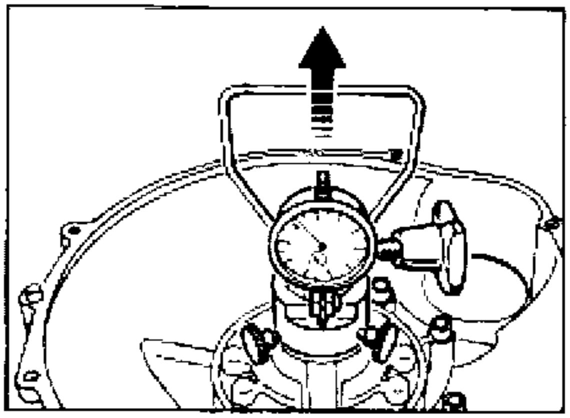
A - Centering pins **9321**

4. Tighten all mounting screws to 46 Nm (34 ftlb).
5. Mount measuring sleeve of Special Tool **9338** with three mounting screws on the stator shaft in such a manner that no play remains.



962-38

6. Slide measuring device of Special Tool **9338** over drive shaft teeth and tighten with clamping screw so that it cannot be tilted.



963-38

7. Set measuring sleeve to zero with a certain preload.

8. Determine end clearance by pulling the handle. Repeat measurement several times.

Note

If a deviation is detected, remove final drive again and fit a thinner or thicker adjusting washer as required.
Recheck end clearance afterwards.

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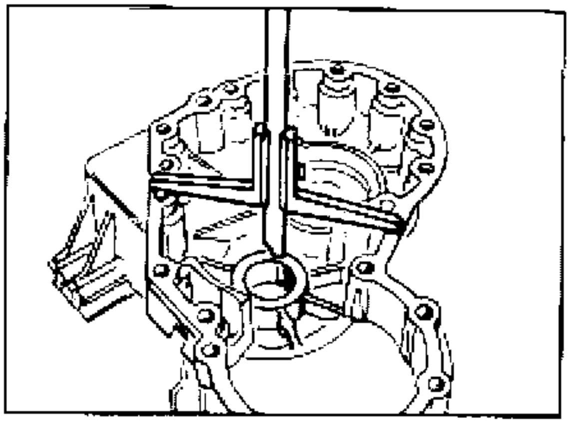
38 91 05 Adjusting the preload of the spur gear taper roller bearings

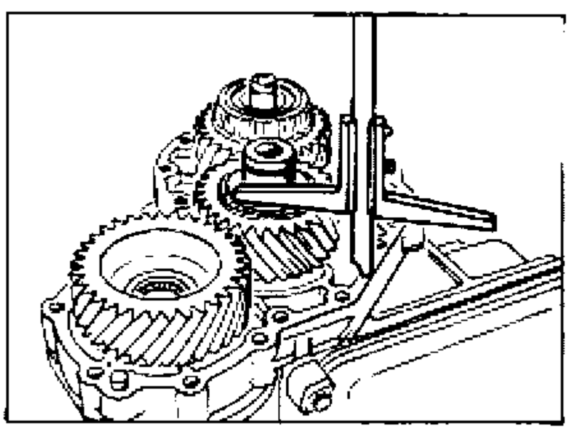
Adjusting spur gear II (intermediate gear)

1. Determine and record dimension "a" (e.g. 45.90 mm).



1030 - 39

2. Fit complete spur gear II and determine dimension "b" (e.g. 44.65 mm). Record measured value.



1031 - 39

3. Determine shim thickness by deducting dimension "b" from dimension "a" and adding a constant value of 0.22 mm (preload and thickness of compressed gasket).

Note

The constant value is 0.22 mm for all transmissions.

Example

Dimension "a" e.g.	45.90 mm
Dimension "b" e.g.	– 44.65 mm
	1.25 mm
Constant	+ 0.22 mm
Shim thickness	1.47 mm

Note

Always round shim thickness up or down to the nearest 0.05 mm figure.

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Adjusting spur gear I (drive gear)

1. Remove intermediate plate.

2. Remove adjusting shim (refer to page 37 - 108, No. 16).

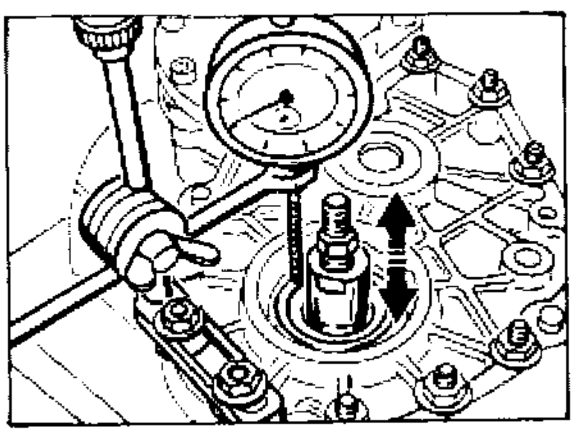
3. Remove spur gear II (intermediate gear) for measurement.

4. Bolt up cover with intermediate plate (but without gasket).
Tighten all screws and bolts to 23 Nm (17 ftlb).

5. Internal puller (e.g. Schrem 20 - 30) into spur gear teeth.

6. Fit dial gauge holder VW 387 with dial gauge and zero out dial gauge with a preload of 2 mm.

7. Move spur gear I up and down on internal puller and read off play on dial gauge (e.g. 1.18 mm).



1029 - 39

A = Dial gauge extension
B = Internal puller

Note

Do no turn or tilt the spur gear when measuring the clearance. Repeat measuring process several times.

8. Determine shim thickness. Measured value plus 0.20 mm (preload and thickness of compressed gasket) equals shim thickness

Note

The constant value is 0.20 mm for all transmissions.

Example

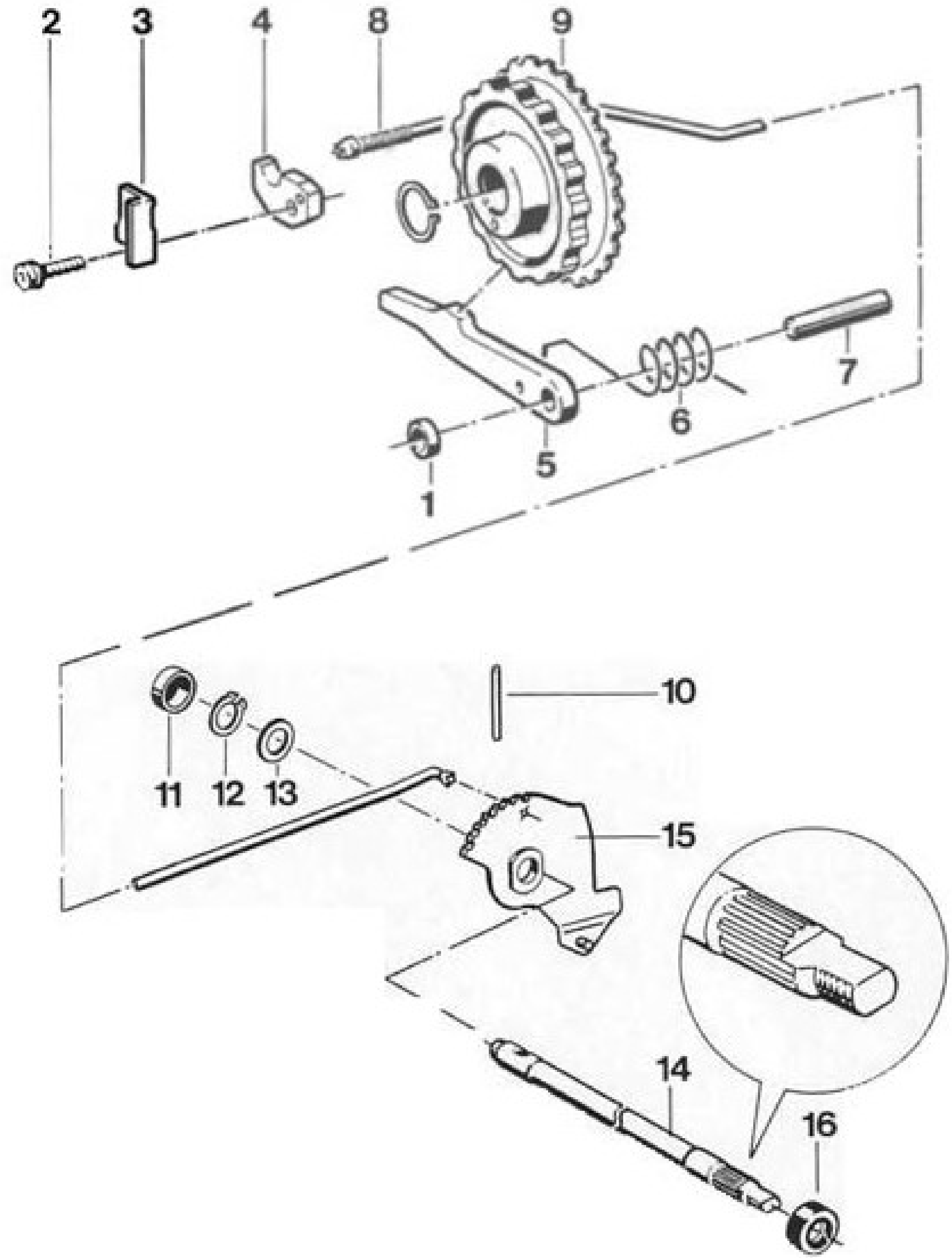
Measured value		1.18 mm
Constant value	+	0.20 mm
Shim thickness		1.38 mm

9. Fit a shim of the determined thickness (1.38 mm in our example).

Note

Always round shim thickness up or down to the nearest 0.05 mm figure.

38 74 37 Disassembling and assembling the parking lock



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No.	Designation	Qty.	Note: Removal	Installation (= ftlb)
1	Bushing	1		
2	Fillister head screw	1	Remove with Torx insert T27	Tighten with 10 Nm
3	Guide plate	1		
4	Guide piece	1		
5	Catch	1		
6	Spring	1		Insert in correct position
7	Pin	1		
8	Circlip	1		
9	Parking lock wheel	1		Make sure that the square ring is correctly seated
10	Clamping pin	1		
11	Protective cap	1	Drive out to the outside by gentle blows on shaft No. 14	Replace
12	Circlip	1		
13	Shim ring	1		
14	Shaft	1	Mark installation position	Insert in correct position
15	Notched disk	1		
16	Rotary shaft seal	1	With transmission in built-in condition, remove multi-functional switch and carefully lever out with a suitable screwdriver	Wind plastic insulating tape around the shaft toothing as an assembly aid, slightly oil sealing lip and press in flush

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Work instructions for disassembling and assembling

Disassembling

1. Remove transmission.

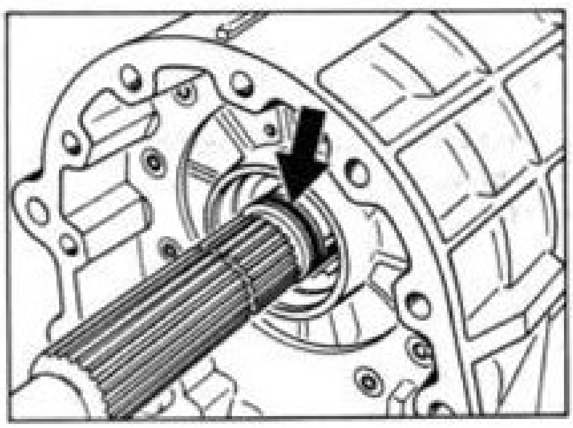
2. Remove intermediate plate.

Note

Part Nos. 11 to 15 can be removed only when the final drive and the ATF pan have been dismantled.

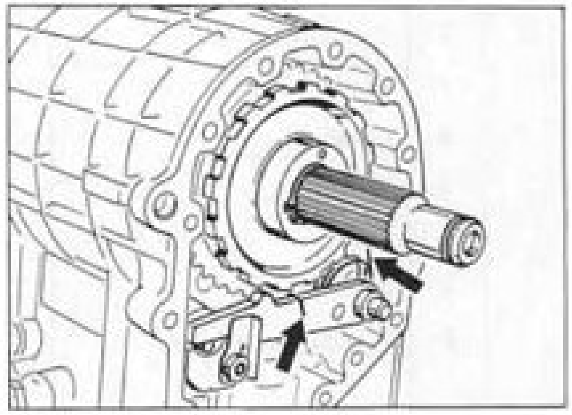
Assembly

1. Replace O-ring on output shaft.



417-37

2. Insert spring for catch in correct position.



418-37

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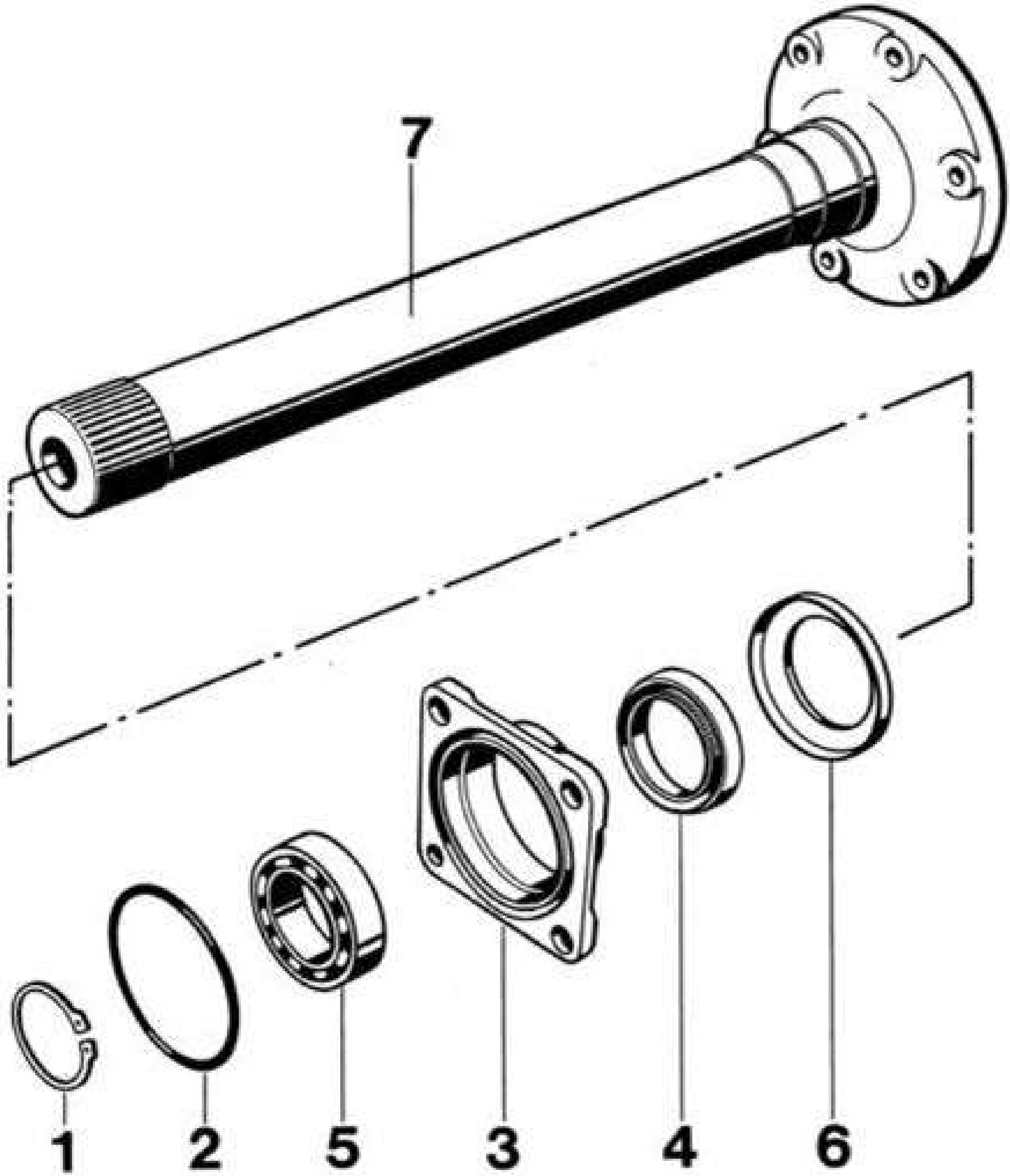
39 25 37 Disassembling and assembling the long joint flange



No.	Designation	Qty.	Note: Removal	Installation
1	Circlip	1		
2	O-ring	1		Replace
3	Bearing cap	1		
4	Rotary shaft seal	1		Pack space between dust lip and sealing lip with grease (e.g. Shell 8420) and press in flush with suitable thrust piece
5	Ball bearing	1	Press off jointly with bearing cap	Press in until it bottoms with a suitable pressure piece
6	Protective ring	1		
7	Joint flange	1		

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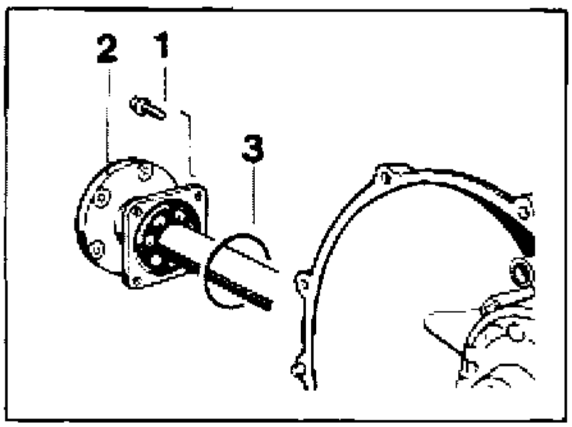
39 25 19 Removing and installing long halfshaft flange

With transmission installed

Removal

1. Remove drive shaft (see page 42 - 17).

2. Unscrew pan head screws of halfshaft flange using special tool **9330** and pull out flange.



2184-39

1 = pan head screw
2 = halfshaft flange
3 = O-ring

Installation

1. Install the parts in the reverse order of dismantling

2. Replace O-ring. Slightly oil new O-ring.

3. Tighten pan head screws to **23 Nm** (17 ftlb).

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39 26 19 Removing and installing short halfshaft flange

With transmission installed

Removal

Note

The halfshaft flange can only be removed if the suspension is partially disconnected.

1. Remove right rear wheel, engine / transmission guard and rear underside panel.

2. Remove heating pipe.

3. Remove control arm cover.

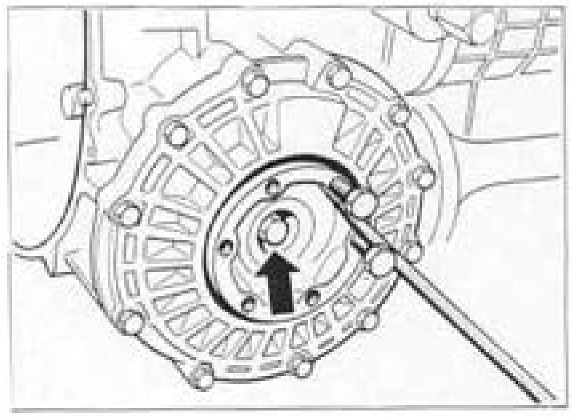
4. Disconnect drive shaft from transmission flange.

5. Partially disconnect suspension (see page 42 - 17 from point 4).

Note

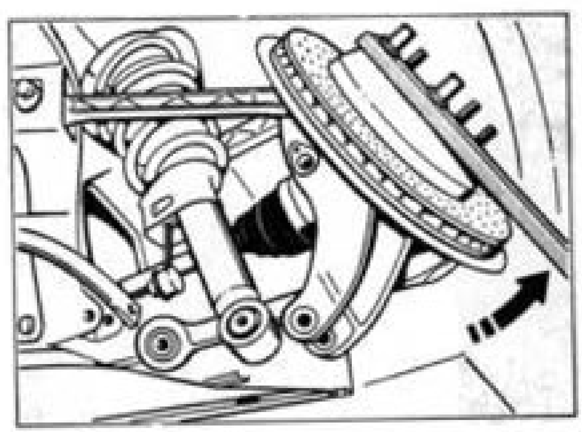
The drive shaft need not be disconnected from the wheel.

6. Unscrew fastening bolt for halfshaft flange (the illustration shows the transmission removed).



408-39

7. Lift wheel carrier using a suitable lever and take out flange (2nd person required).



2065-42

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Installation

1. Install the parts in the reverse order of removal.

2. Tighten screws to specified torque figures.

Printed in Germany 12, 1995

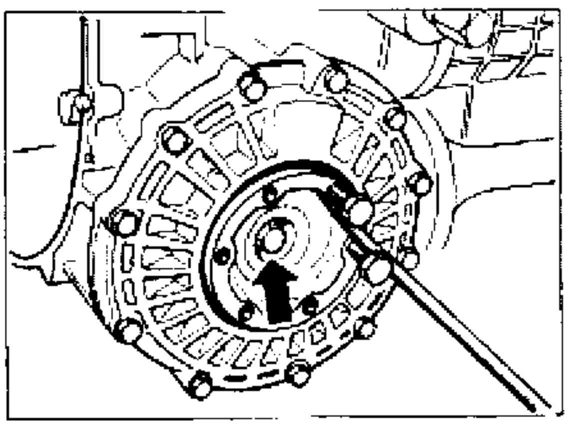
39 22 19 Removing and installing the rotary shaft seal for the short joint flange

Removing

Note

The rotary shaft seal can also be replaced with the transmission in built-in condition.

1. Unscrew the hexagon screw for the joint flange and remove flange.



408-39

2. Lever out rotary shaft seal with VW 681

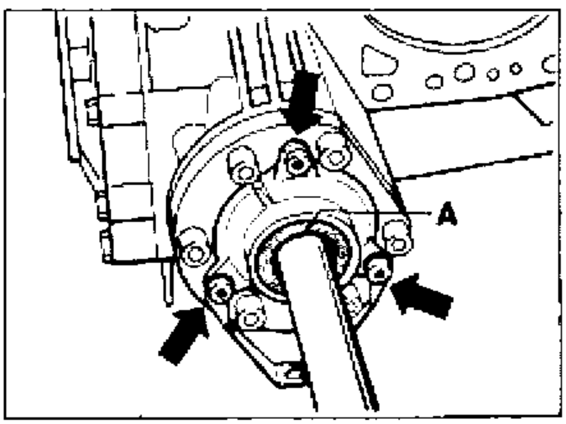
Installing

1. Press in rotary shaft seal flush with suitable pressure piece.

2. Tighten hexagon screw for joint flange with **46 Nm (34 ftlb).**

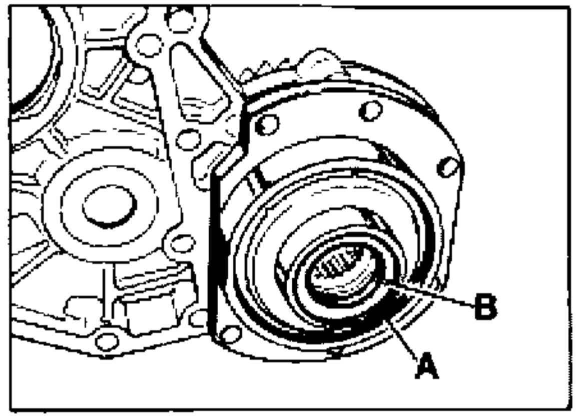
39 28 19 Removing and installing drive pinion seal rings

1. Drain ATF
2. Remove engine with gearbox (do not unbolt engine from gearbox.
3. Remove intermediate plate (see page 37 - 125).
4. Remove bearing cap with **Torx insert T 27** and replace rotary shaft seal. Pack space between dust lip and sealing lip with grease (e.g. Shell 8420).
   Press-in depth 2.0 ± 0.5 mm.



A = Rotary shaft seal 422-39

5. Fit bearing cap with new O-ring and tighten fixing screws with **10 Nm (7.5 ftlb)**.
6. Remove snap ring for bearing cap at intermediate plate and pull out cap.



423-39

A = Snap ring

B = Rotary shaft seal

7. Replace rotary shaft seal.
   Pack space between dust lip and sealing lip with grease (e.g. Shell 8420).
   Press-in depth 2.0 ± 0.5 mm
8. Fit new O-ring and wet with ATF.
9. Insert bearing cap and fit snap ring.
10. Fill with ATF.

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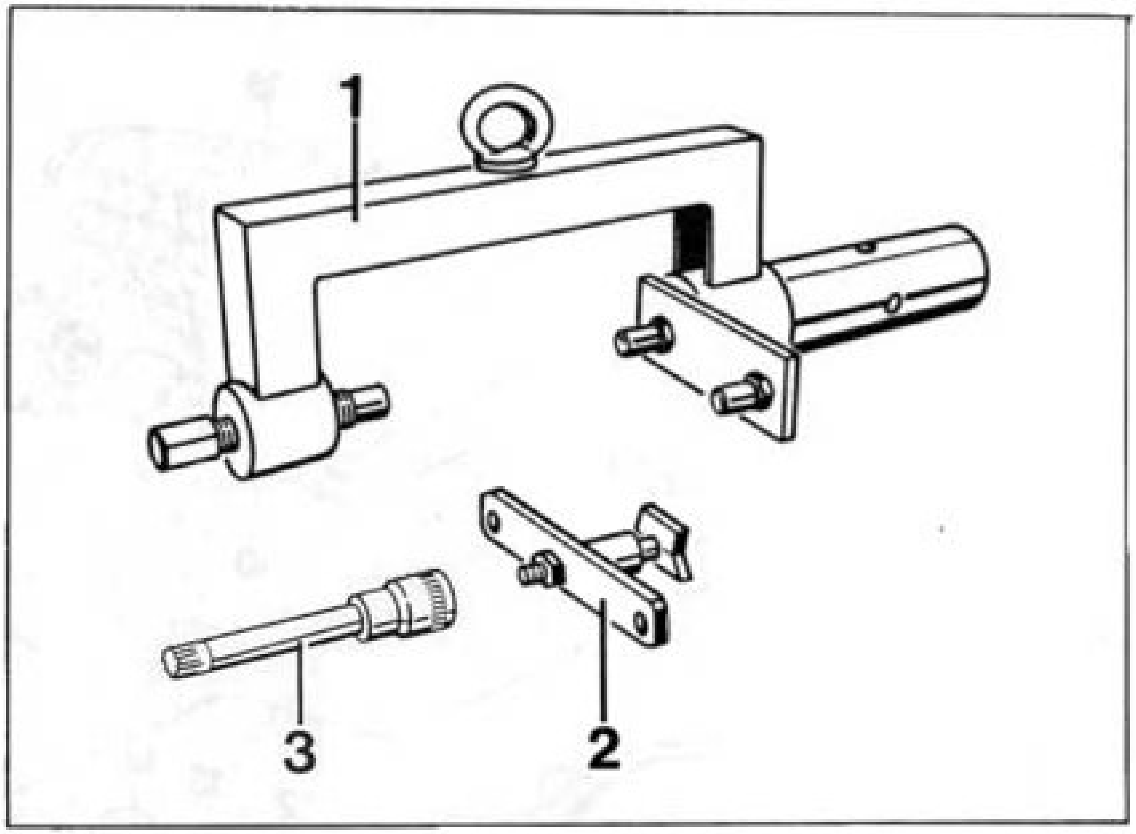
39 01 19 Removing and installing rear transmission case

Tools



No.	Designation	Special tool	Order number	Explanation
1	Transmission holder	9324	000.721.932.40	
2	Holding device	9325	000.721.932.50	
3	socket key insert	9330	000.721.933.00	

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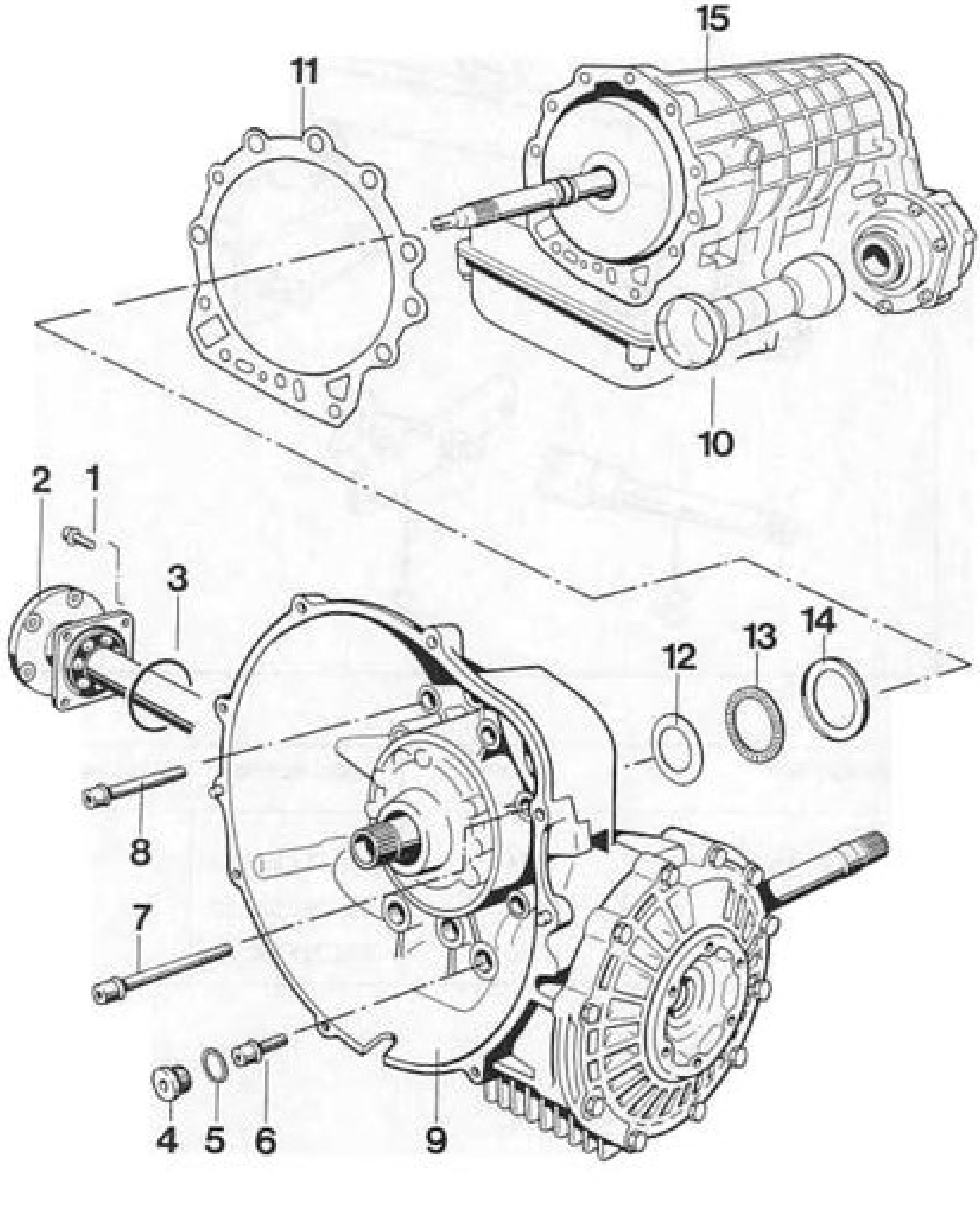
39 01 19 Removing and installing final drive



424-37

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No.	Designation	Qty.	Note: Removal	Installation (= ftlb)
1	Fillister head screw	4	Use Special Tool 9330 to remove	Tighten witht 23 Nm (17)
2	Joint flange	1		
3	O-ring	1		Replace, oil slightly
4	Screw plug	3	Remove with Torx insert T55	Tighten with 50 Nm (36)
5	Sealing ring	3		Replace
6	Fillister head screw	3	Remove with Torx insert T50	Tighten with 46 Nm (34)
7	Fillister head screw	3	Remove with Torx insert T50	Tighten with 46 Nm (34)
8	Fillister head screw	6	Remove with Torx insert T50	Tighten with 46 Nm (34)
9	Rear transmission case	1		
10	Protective tube	1		Large diameter to final drive
11	Seal	1		Replace, glue to case with a small quantity of grease (vaseline)
12	Shim*	X	Note thickness for re-installation	Rederternine thickness if required (refer to page 38 - 171)
13	Needle cage	1		
14	Angle ring	1		Fit in correct position
15	Automatic transmission	1		Pay attention to installation position of clutch A. Make sure that the square drive shaft rings are correctly fitted. Coat the drive shaft with a small quantity of grease in the area of the square rings.

* Must be checked and, if required, corrected after each transmission installation.

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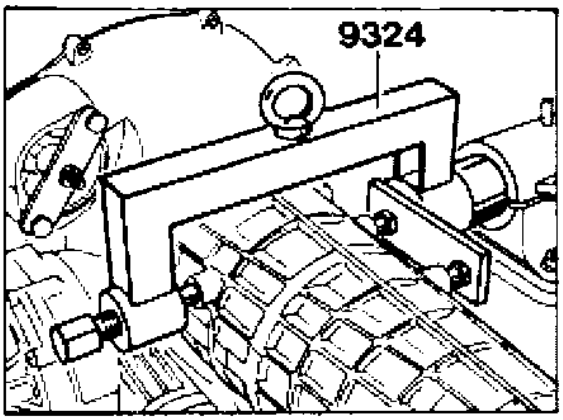
Assembly instructions for removal and installation

Removal

1. Remove the transmission and torque converter.

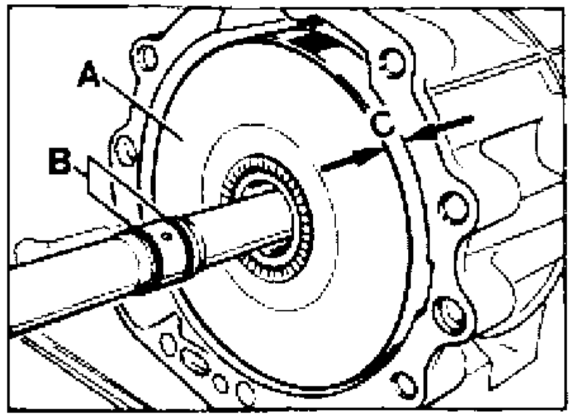
2. Secure transmission on assembly block with holding device **9324**.



416-37

Installation

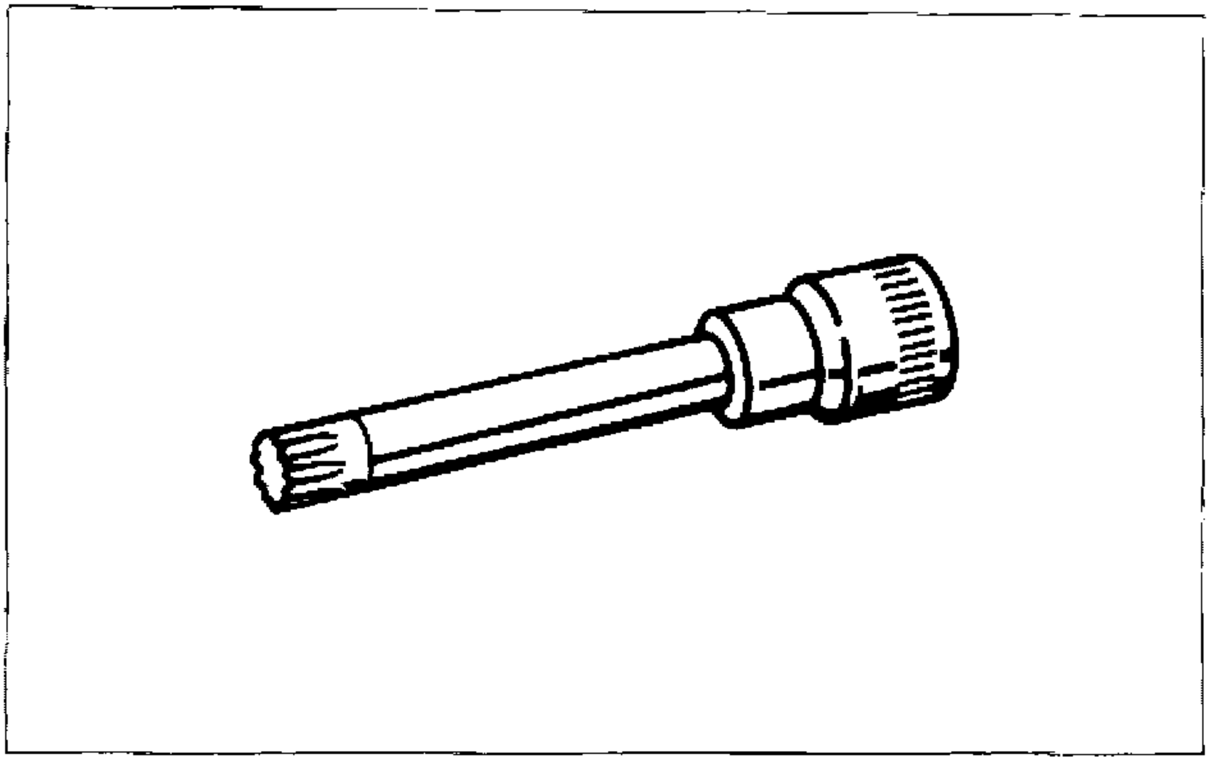
1. Check the installation position of clutch A. The installation depth has been reached when the distance **"C" is approx. 8.5 mm.**



416-37

A = Clutch A
B = Square rings
C = Installation depths

39 09 19 Removing and installing differential and drive pinion



No.	Designation	Special tool	Order number	Explanation
	Socket	9330	000.721.933.00	

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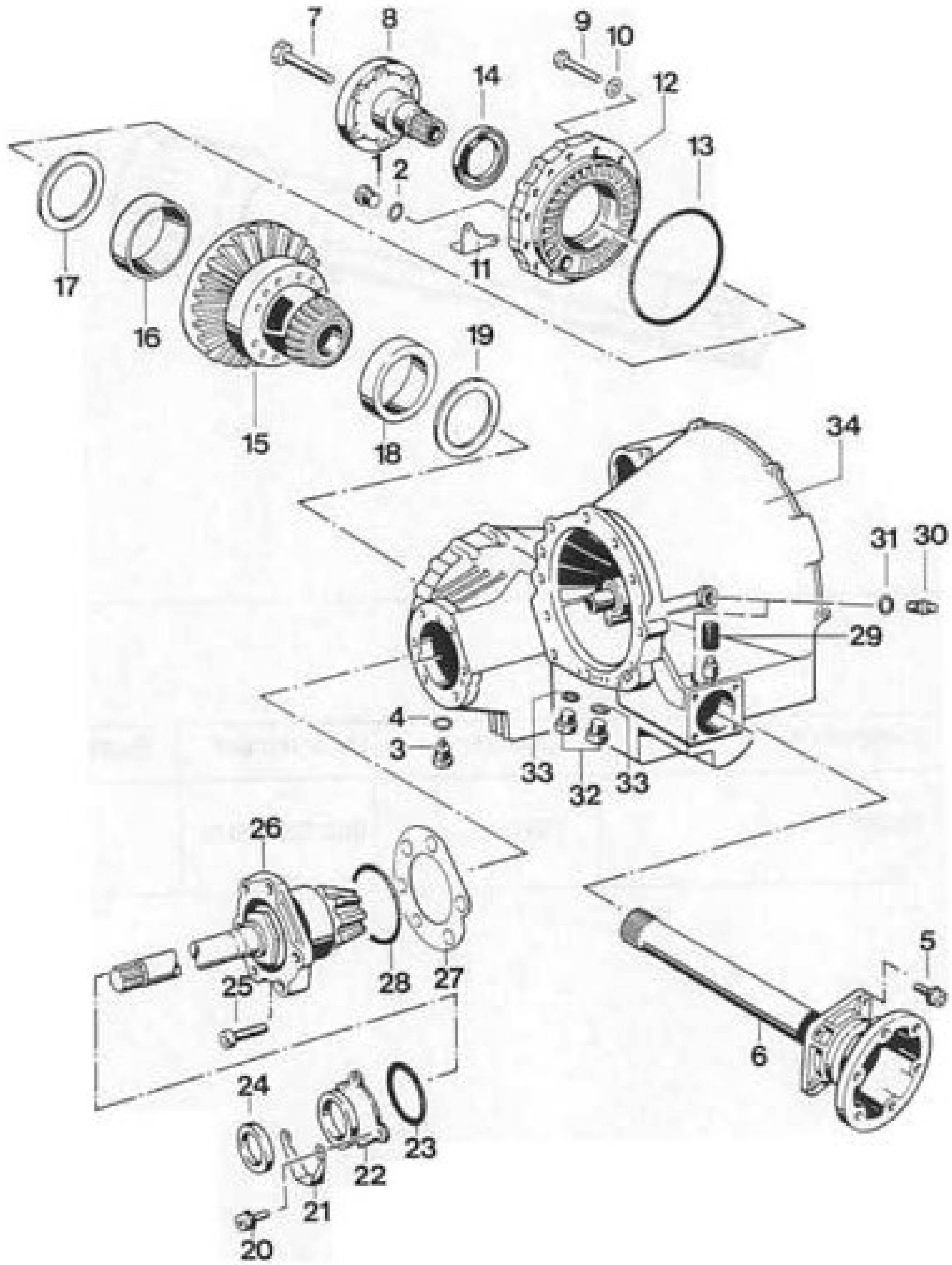
39 09 19 Removing and installing differential and drive pinion



946-39

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No.	Designation	Qty.	Note:	
			Removal	Installation
1	Plug	1		Tighten to 50 Nm (37 ftlb)
2	Seal	1		Replace
3	Plug with solenoid	1		Tighten to 50 Nm (37 ftlb)
4	Seal	1		Replace
5	Pan head screw	4	Remove with Special Tool 9330	Tighten to 23 Nm (17 ftlb)
6	Halfshaft flange	1		For dismantling, refer to page 39 - 102
7	Hexagon head bolt	1		Tighten to 46 Nm (34 ftlb)
8	Halfshaft flange	1		
9	Hexagon head bolt	11		Tighten to 23 Nm (17 ftlb)
10	Spring washer	11		
	Bracket	1		
12	Transmission side cover	1		
13	Round seal	1		Replace, oil lightly
14	Shaft seal	1	Replace	Pack space between dust lip and sealing lip with grease (e.g. Shell 8420) and press in flush
15	Differential	1		
16	Taper roller bearing outer race	1		
7	Adjusting shim "S_1"	X	Record thickness for refitting	Redetermine thickness if required
18	Taper roller bearing outer race	1		
19	Adjusting shim "S_2"	X	Record thickness for refitting	Redetermine thickness if required
20	Pan head screw	3		Tighten to 10 Nm (7 ftlb)

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No.	Designation	Qty.	Note: Removal	Installation
21	Bracket	1		
22	Beanng cover	1		
23	Round seal	1		Replace
24	Shaft seal	1		Replace. Pack space between dust lip and sealing lip with grease (e.g. Shell 8420). Press in to depth of 2.0 ± 0.5 mm
25	Pan head screw	6		Tighten to 50 Nm (37 ftlb)
26	Drive pinionm with bearing assembly	1	To drive out, use a plastic hammer to apply light blows on the drive pinion head	Observe matching number, readjust if required
27	Adjusting shim "S3"	X	Record thickness for reassembly	Redetermine thickness if required
28	Round seal	1		Replace, oil lightly
29	Breather	1		
30	Screw-in flange	2		Tighten to 35 Nm (26 ftlb)
31	Seal	2		Replace
32	Plug	2		Tighten to 25 Nm (18 ftlb)
33	Seal	2		Replace
34	Final drive	1		

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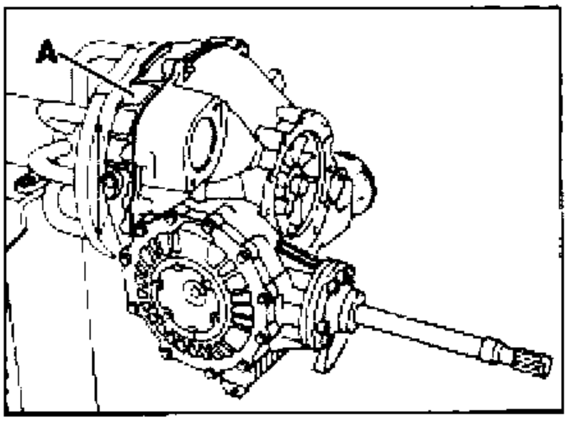
Removal note

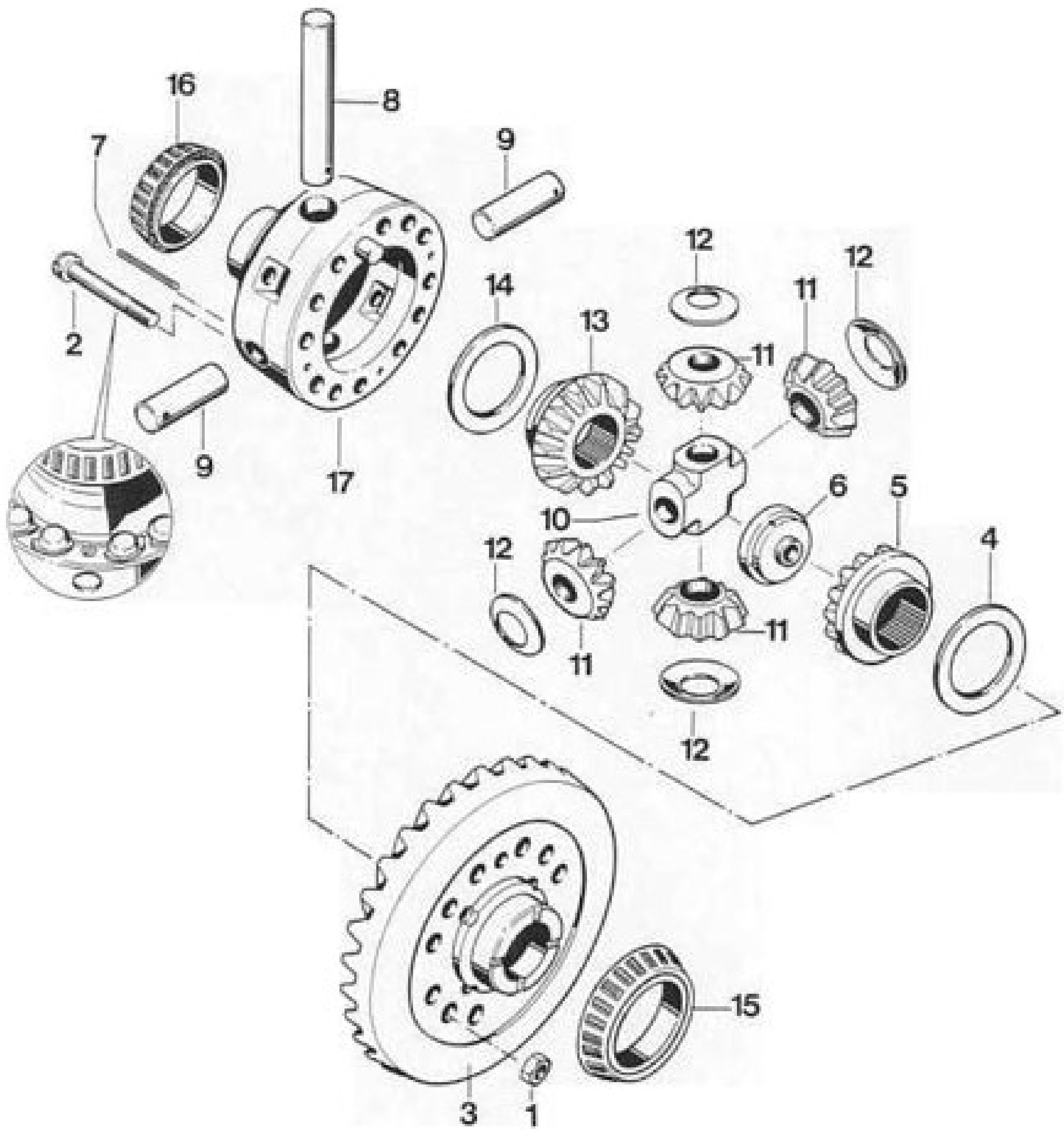
Mount rear transmission case with converter housing on assembly support.



965-39

A - Converter housing (spare part)

39 09 37 Dismantling and assembling differential



960-39

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No.	Designation	Qty.	Note: Removal	Installation
1	Hexagon nut	12		replace. Thread must be dry and free from grease. Secure with Loctite 262. Tightening torque: 85 Nm (62.7 ftlb)
2	Lock screw	12		replace. Thread must be dry and free from grease. Install in correct position.
3	Ring gear	1	Separate from housing by applying light blows with a plastic hammer	Observe matching number. Readjust if required
4	Thrust washer	1		
5	Shaft bevel gear	1		
6	Nut	1		
7	Roll pin	3		Press in place in correct position
8	Stud (long)	1		Lock with roll pin
9	Stud (short)	2		Lock with roll pin
10	Cross fitting	1		
11	Bevel gear	4		
12	Thrust washer	4		
13	Shaft bevel gear	1		
14	Thrust washer	1		
15	Taper roller bearing inner race	1	Pull of with suitable puller, or remove ring gear and take out from inside through bores	Heat to approx. 120° C and press on
16	Taper roller bearing inner race	1	Pull off with P 263	Heat to approx. 120° C and press on
17	Housing	1		Fit centering pin with Loctite 262

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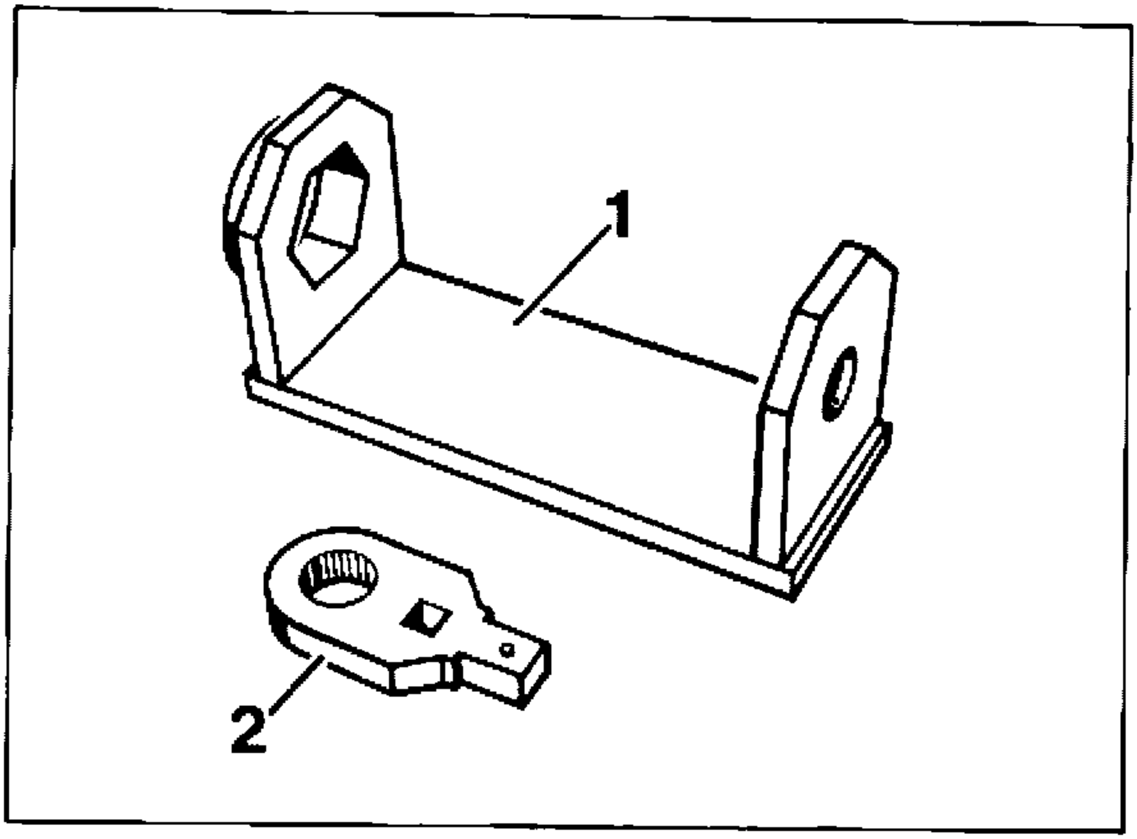
39 30 19 Dismantling and assembling drive pinion



967-39

No.	Designation	Special tool	Order number	Explanation
1	Retainer bracket	9337	000.721.933.70	
2	Insert	9282	000.721.928.20	

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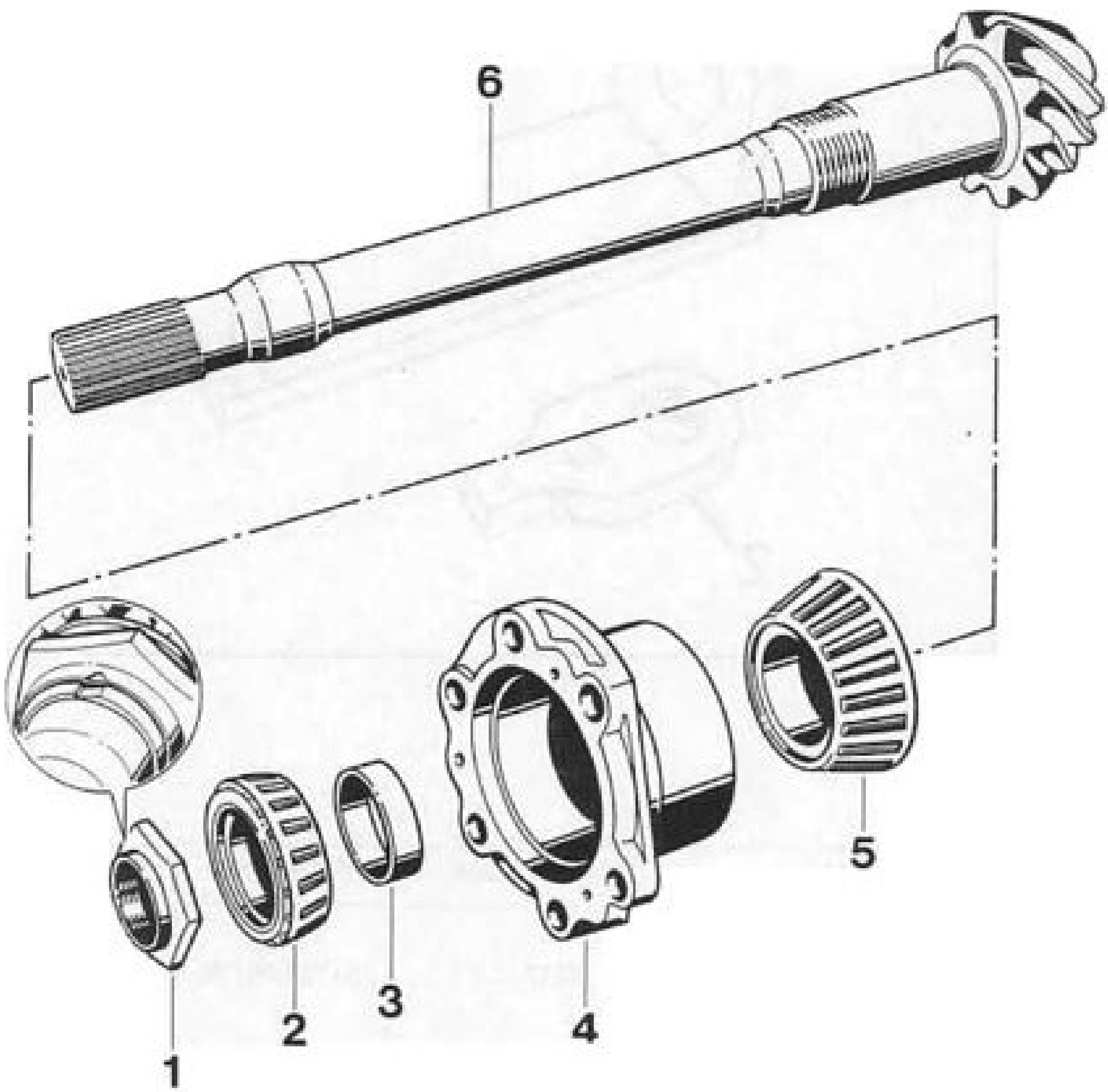
39 30 19 Dismantling and assembling drive pinion



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No.	Designation	Qty.	Note: Removal	Installation
1	Lock nut	1	Undo with Special Tool **9337** and **9282**	Tighten to 250 Nm (184 ftlb) and secure by upsetting the flange in two places
2	Taper roller bearing inner race	1	Press off with bearing flange	Heat to approx. 120° C and press on
3	Adjusting ring	X		
4	Bearing flange	1		Only available pread-justed as a complete spare assembly (items 2 to 5)
5	Taper roller bearing inner race	1	Press off with separating device	Heat to approx. 120 °C and press on
6	Drive pinion	1		Readjust if required, observe matching number

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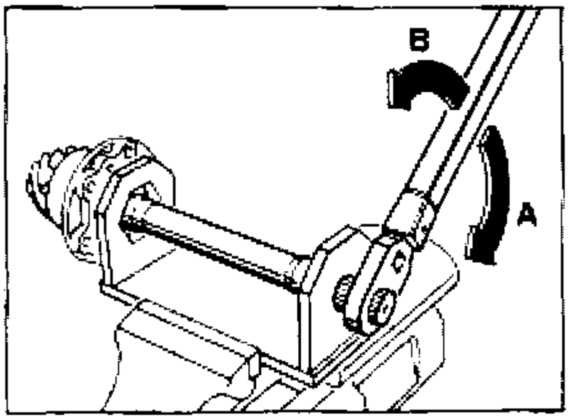
Notes on removal

Dismantling

Undo lock nut with Special Tools **9337** and **9282**.

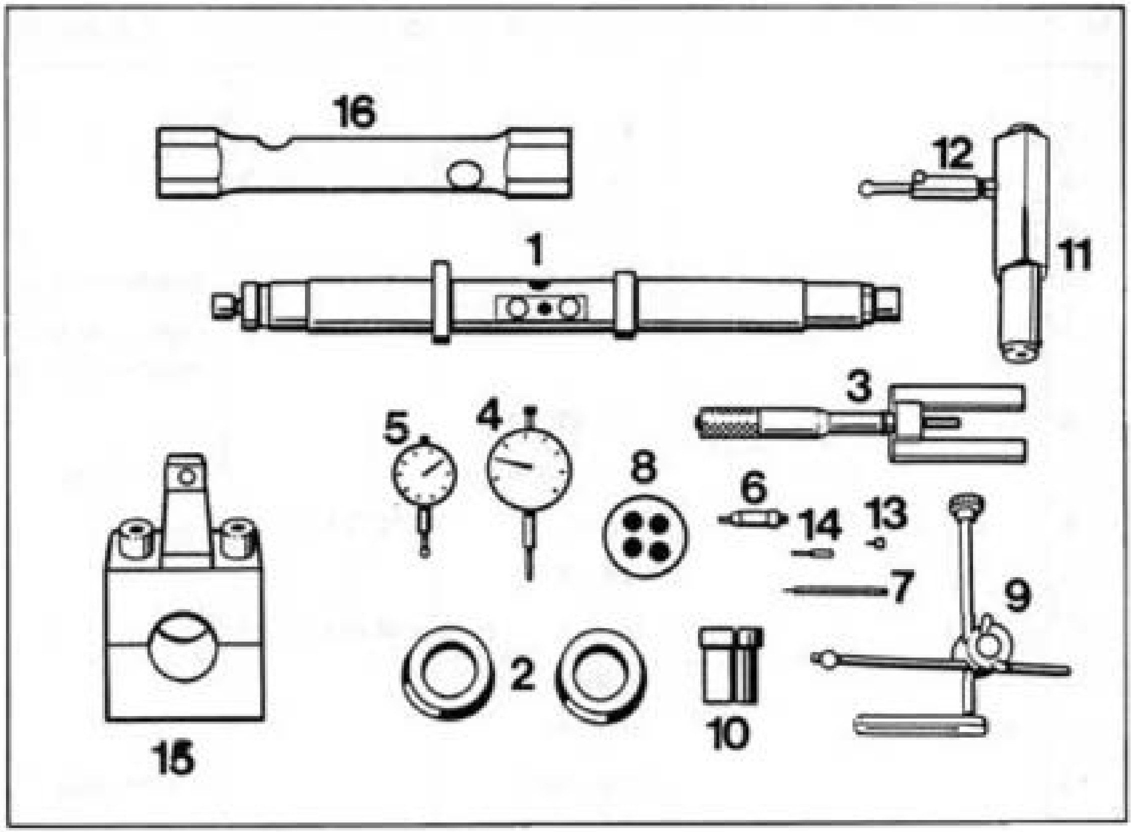


968-39

A - Undoing
B - Tightening

39 08 15 Adjusting drive set

Tools



Adjusting drive set

Tools

No.	Designation	Special tool	Order number	Explanation
1	Measuring mandrel	VW 385/1		
2	Centering discs	9327	000.721.932.70	
3	Master gauge	VW 385/30		
4	Dial gauge	–		commercially available
5	Dial gauge	–		commercially available, measuring range 3mm
6	Gauge plunger	VW 385/14		
7	Dial gauge extension	VW 385/56		30 mm long
8	Gauge block plate	9281	000.721.928.10	
9	Dial gauge bracket	VW 387		
10	Clamping sleeve	9145	000.721.914 50	
11	Adjusting device	VW 521		
12	Measuring lever	VW 388		
13	Dial gauge extension	VW 382/10		6.0 mm long
14	Dial gauge extension	VW 385/53		14 mm long
15	Clamping device	9339	000.721.933.90	
16	Socket wrench (24 mm A/F)	–		commercially available (e.g. Stahlwille No. 10 750)

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Practical procedure when readjusting the drive set

If it becomes necessary to adjust drive pinion and ring gear, follow the below sequence to ensure an efficient working procedure:

1. Determine the total shim thickness "Stot" (S_1 plus S_2) for the specified preload on the taper roller bearings/differential.

2. Determine the thickness of shim "S_3".

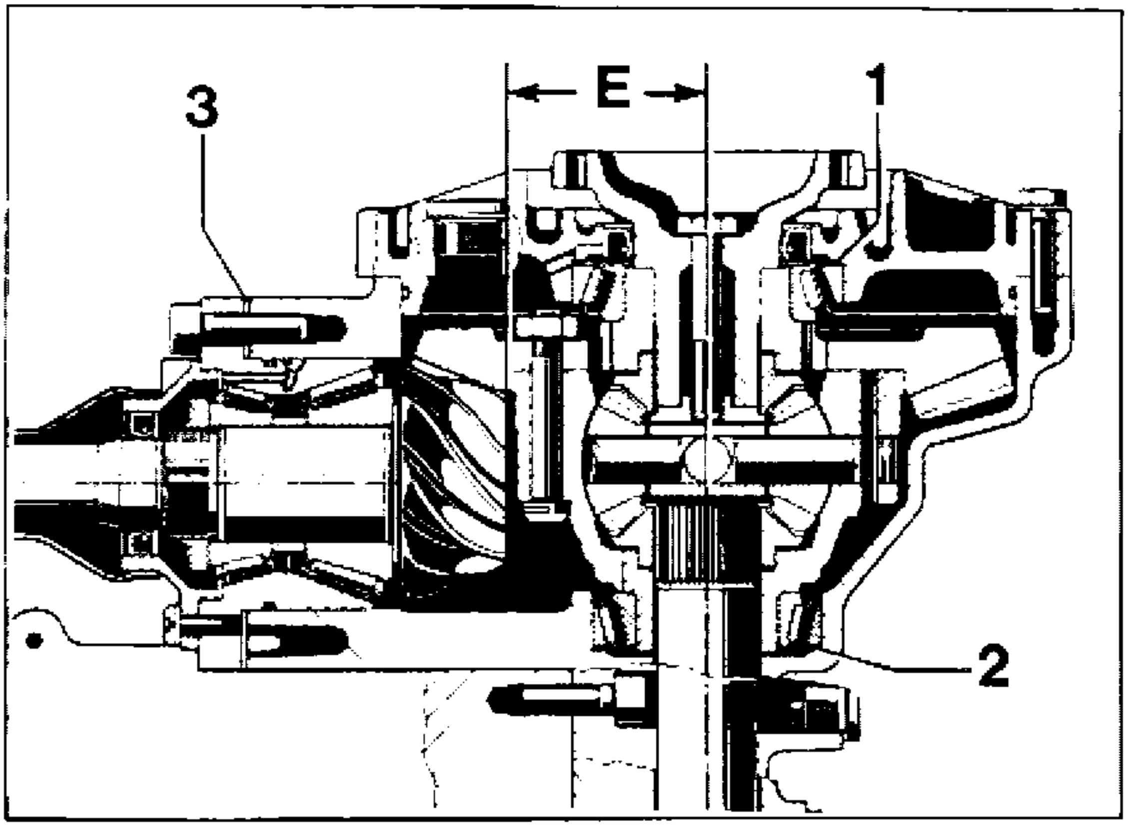
3. Split the total shim thickness "Stot" into S_1 and S_2 so that the specified circumferential backlash is present between ring gear and drive pinion.

The aim of this adjustment is to restore the smoothest running position which has been achieved on test equipment in the production line.

To achieve correct results, greatest possible cleanliness for all assembly work and measuring procedures is essential.

When assembling the final drive assembly, it is only necessary to readjust drive pinion and ring gear or drive set if components have been replaced which have a direct influence on the adjustment. Refer to the following table to avoid unnecessary adjustment procedures!

Adjust: / Replaced component	Ring gear (S_1 + S_2)	Drive pinion (S_3)
Rear transmission case	x	x
Transmission side cover	x	
Bearing assembly for drive pinion	x	x
Drive set	x	x
Differential housing	x	
Taper roller bearing for differential	x	



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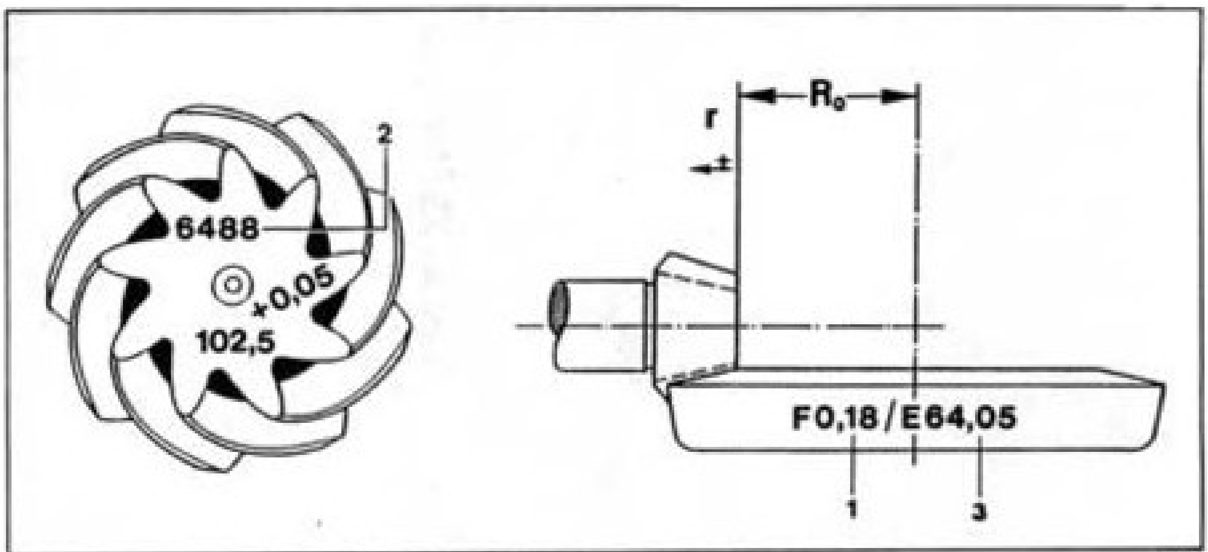
1 – Spacer S_1
2 – Spacer S_2
3 – Adjusting shim S_3
E – Adjustment dimension

Correct results can only be achieved if assembly work and measuring procedures are carried out carefully and with maximum cleanliness.

Adjusting drive set

General

The setting of drive pinion and ring gear is a determining factor for the service life and smooth running of the rear-axle drive. Drive pinions and ring gear that have been checked for good tooth contact pattern and low noise in both directions of rotation on special test equipment are therefore matched during production. The position at which smoothest running can be achieved is determined by shifting the drive pinion axially, with the ring gear being kept within the tolerance of the prescribed tooth backlash. The deviation "r" from the specified design dimension "Ro" is measured, added to the design dimension "Ro" and engraved on the ring gear as setting value "E".



Ro - design dimension (64.00 mm)
r - deviation r
1 - Backlash "F" (e.g. 0.18 mm)
2 - Matching number
3 - Setting value "E" (Ro + r)

All other characters are used for adjustment in production.

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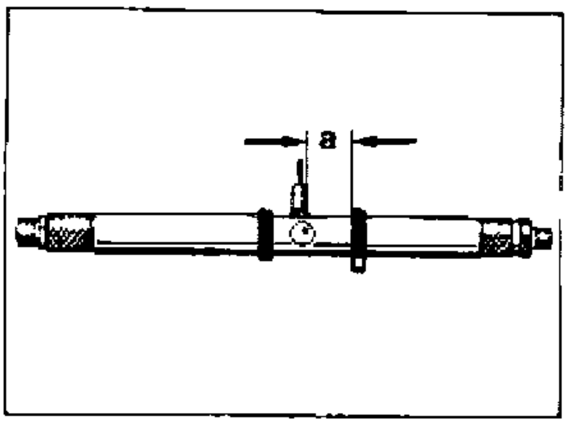
Adjusting drive pinion

Note

The setting dimension "E" is indicated on the ring gear.

1. Install drive pinion **without** shims "S_3" and tighten all pan head screws of bearing assembly to 50 Nm (37 ftlb).

2. Rotate adjustable stop ring along with spindle towards measuring plunger as far as it will go and set second setting ring to dimension"a".



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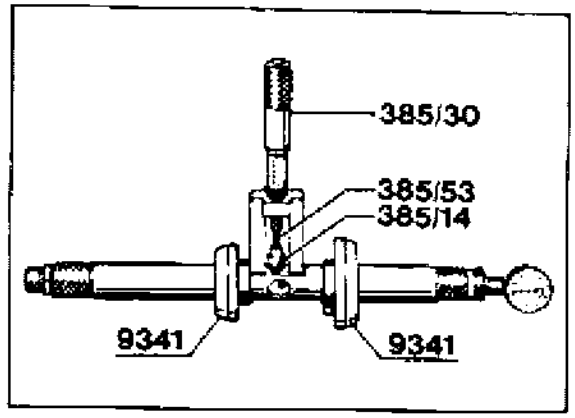
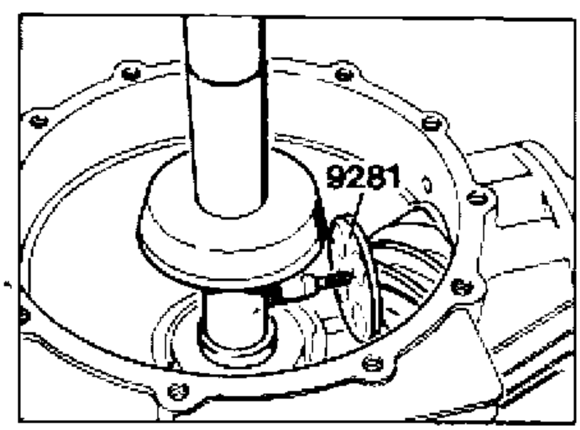
a = 30 mm

3. Assemble measuring mandrel and set with master gauge **VW 385/30** to setting dimension "E" (64.05 mm in the example). Set dial gauge (3 mm measuring range) to zero with 1 mm preload.



4. Put gauge block plate **9281** on drive pinion head and insert measuring mandrel with dial gauge towards transmission side cover into transmission case. Dial gauge extension points towards center of drive pinion.



5. Fit transmission side cover without shaft seal and sealing ring and tighten crosswise with 4 hexagon head bolts.

Note

Do not use a hammer when fitting the transmission side cover (the gauge block plate held by magnets might fall off). Locate cover in correct installation position only by tightening the hexagon head bolts uniformly.

6. Using a socket wrench (24 mm A/F, e.g. Stahlwille No. 10 750), pull adjustable centering disc over spindle towards the outside until the measuring mandrel can just about be turned.

7. Turn measuring mandrel carefully until the dial gauge extension is vertical to the face of the drive pinion head. At this point, the pointer of the dial gauge is at its maximum deflection (reversing point) and the dial gauge must now be read.

Note

The measured value always deviates from the set dimension in clockwise direction (small pointer on the dial gauge is between 1 and 3), i.e. if the dial gauge is set with a preload of 1 mm, the value deviating from 1 is taken as the shim thickness "S_3" to be inserted.
Always round up or down to the nearest 0.05 mm (e.g. 1.63 to 1.60 mm).

8. After inserting the required shims, check the setting value "E" once again. A deviation of ± 0.03 mm is permissible.

Adjusting ring gear

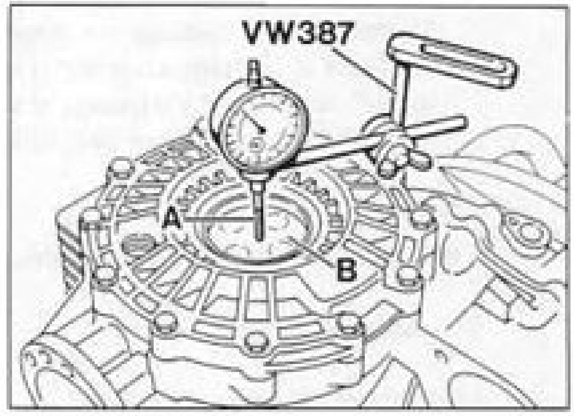
Determining total shim thickness "Stot." ($S_1 + S_2$)

The ring gear must be readjusted if the transmission case, transmission side cover, taper roller bearing for differential, differential housing or drive set have been replaced.

Note

The drive pinion must be removed in order to determine the preload of the differential tapered roller bearings.

1. Remove adjusting shim "S_1" (in transmission side cover).
2. Adjusting shim "S_2" remains in the transmission case.
3. Make sure that the bearing outer races of the tapered roller bearings are well seated in the transmission case or in the transmission side cover.
4. Insert differential into transmission case and rotate several times.
5. Fit transmission side cover without seals and tighten all hexagon head bolts to **23 Nm (17 ftlb)**.
6. Place gauge block plate **9281** on the collar of the differential.
7. Fasten universal dial gauge holder **VW 387** with dial gauge and extension to the case and set to 0 with 2 mm preload.



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A - Dial gauge extension (approx. 30 mm long)
B - Gauge block plate 9281

8. Using a suitable tool, move differential up an down.
   Read off backlash on the dial gauge and note.

Note

Do not turn differential while measuring backlash as this will give an incorrect reading.

9. Remove adjusting shim "S_2" and determine thickness using a micrometer.
10. Calculate "Stot".
    "Stot." = Thickness of adjusting shim "S_2"
    + measured value
    + pressure fit of taper roller bearings

Example

Thickness of adj. shim "S_2"	1.70 mm
Measured value	0.91 mm
Press fit (constant value)	0.24 mm
"Stot"	2.85 mm

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11. Spread calculated shim thickness "Stot" as follows.
    To start with the backlash adjustment, the thickness of adjusting shim "S_1" is reduced by 0.40 mm while the thickness of adjusting shim "S_2" is increased by 0.40 mm.

Example

Total shim thickness of adjusting shims

$S_1 + S_2 = 2.85$ mm

Thickness of adjusting shim "S_1"

$$\frac{2.85\ \text{mm}}{2} \qquad \begin{array}{r} 1.425\ \text{mm} \\ -\ 0.40\ \ \text{mm} \\ \hline 1.025\ \text{mm} \end{array}$$

Thickness of adjusting shim "S_2"

$$\frac{2.85\ \text{mm}}{2} = \qquad \begin{array}{r} 1.425\ \text{mm} \\ +\ 0.40\ \ \text{mm} \\ \hline 1.825\ \text{mm} \end{array}$$

Note

The adjusting shims are avail. in thicknesses of 1.0...2.0 mm in increments of 0.05 mm.

The shim tichknesses calculated must be rounded up or down for plausible dimensions that will not alter the total shim thickness S_1 and S_2.

Example

Calculated shim thickness

$S_1 + S_2 = 1.025 + 1.825 = 2.85$ mm

Rounded down shim thickness

$S_1 + S_2 = 1.00 + 1.85 = 2.85$ mm

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Adjusting circumferential backlash

Note

The backlash to be set is embossed on the ring gear.

1. Drive pinion set using the shims "S_3" determined while adjusting the drive pinion and tighten all mounting screws to 50 Nm (37 ftlb).

Note

Make sure the collar nut of the drive pinion is tightened to 250 Nm (184 ftlb).

2. Fit the adjusting shims (S_1 + S_2) determined into the transmission case and the transmission side cover, respectively.

3. Fit differential and transmission side cover and tighten all hexagon head bolts of the cover to 23 Nm (17 ftlb).

Note

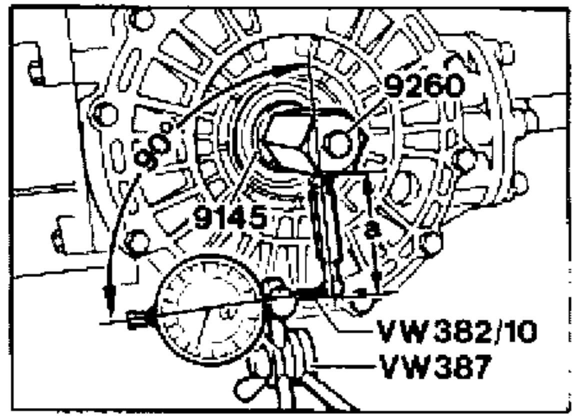
Always make sure that there is a certain amount of backlash when tightening the hex bolts. Never allow the drive pinion to bind.

4. Assemble measuring lever **VW 388** and adjusting device **VW 521/4** and adjust lever length to 80 mm with the plunger. Refer to dimension "a" in the picture.

5. Insert adjusting device with clamping sleeve (Special Tool 9145) into the differential and clamp firmly.

6. Rotate differential in both directions several times to settle the tapered roller bearings.

7. Fit universal dial gauge holder with flat extension in such a way as to produce a right angle between dial gauge axis and lever.



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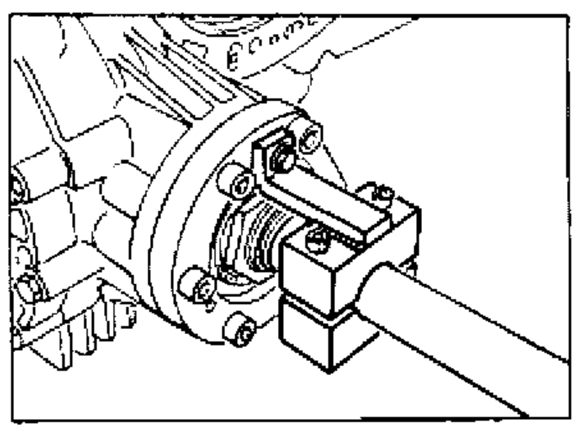
Dimension "a" = approx. 80 mm

8. Turn ring gear carefully at the clamping screw of the adjusting device up to the stop and set the dial gauge to zero. Turn back ring gear and read off circumferential backlash. Note the reading.

Note

When carrying out measurements, the drive pinion must be blocked with Special Tool **9339**.



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9. After turning the ring gear a further 90° each, repeat measuring procedures three times. The measured values must not deviate from one another by more than 0.05 mm.

Note

The backlash to be adjusted is embossed on the ring gear. The actual value may be less than the specified value by - 0.05 mm. Under no circumstances must the backlash be greater than the specified value.

10. If the required backlash cannot be obtained, replace spacers (S1 + S2) again. The total shim thickness (S tot.) must not be altered, however.

Note

Changing the shim thickness of "S1" or "S2" by 0.05 mm shim results in a change of backlash by approx. 0.1 mm.

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